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# Additively manufactured metamaterials for acoustic absorption: a review

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**ABSTRACT**

Noise pollution, a major concern in modern life, contributes to various mental and psychological problems. In this study, we concentrate on the recent developments in metamaterials, particularly additively manufactured acoustic-absorbing metamaterials. These have demonstrated enormous potential for noise reduction across a broad range of frequencies. The current review attempts to classify these additive-manufactured acoustic absorbing metamaterials as perforated, slotted, cellular, and hybrid. Herein, we present a detailed study of the efficiencies and applicability of each subclass, as well as a critique of the experimental research undertaken within each subclass. The review focuses on the subtle features of their operation by evaluating the effect of varying key parameters on the sound absorption capabilities of these metamaterials. Through this comprehensive review, we aim to highlight the current status of this evolving area and motivate more study and innovation in pursuit of effective noise pollution reduction measures.

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Noise pollution; acoustic metamaterials; acoustic absorption; additive manufacturing; 3D printing

## 1. Introduction

Noise pollution has become a serious environmental hazard for people worldwide as industry and transportation operations have expanded. Excessive noise exposure has a deleterious influence on the health of both humans and wildlife. Noise pollution is common in many areas with a human population and is most commonly associated with manufacturing, transportation, and social events. Individuals can be exposed to high levels of noise while at work, school, residence, and various other locations. Although exposure to high noise levels might disrupt everyday tasks and performance, there are also concerns that noise, as an interaction with the environment, may have long-term dangers for human growth and well-being [1]. The sources of noise pollution are diverse, ranging from low to high-frequencies, and include traffic, construction, industrial activities, and even household appliances. The detrimental effects of noise pollution on human wellness and health have been extensively studied, including hearing damage, sleep disturbance, stress, and cardiovascular problems, as shown in Figure 1. As a result, a growing body of research is exploring new and innovative strategies for mitigating noise pollution [2]. Acoustic rehabilitation is one of the methods that can be implemented to improve the acoustic condition of the built environment. However, this method is prohibitively expensive since it involves modification and reconstruction processes; thus, it is not a feasible solution for implementation [3]. An alternative approach, perhaps a more practical and affordable solution to address the acoustical issue in a specific area or room, is installing an acoustic absorber [4]. Acoustic absorbers limit the propagation of acoustic waves along their route by using acoustic absorption, a typical noise reduction technique [5]. Acoustic absorption can be mathematically defined as the ratio of sound energy absorbed by a structural/material system to the incident sound energy in that system, characterised as the sound absorption coefficient.

Acoustic absorption research has traditionally concentrated on conventional methods and materials; however, there is currently a growing interest in metamaterials. Metamaterials are structures engineered artificially and have unique topological characteristics that differ from conventional materials without contravening the laws of physics. Metamaterials evolved from electromagnetics when, in 1968, the theoretical physicist Veselago observed that electromagnetic materials with

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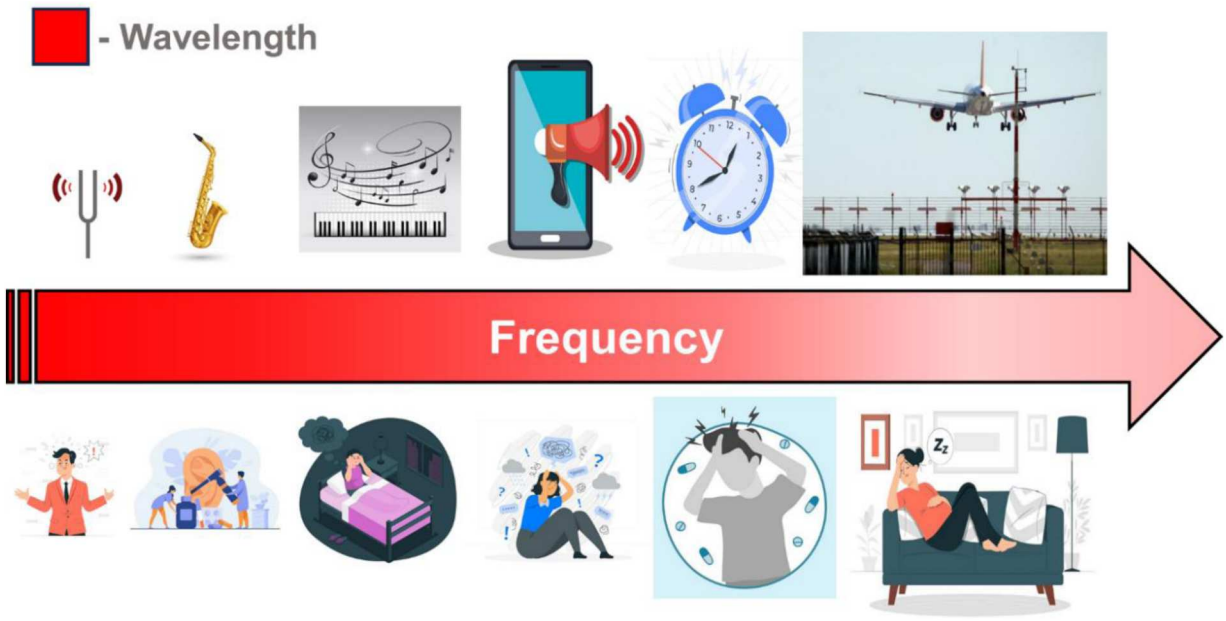



**Figure 1.** Sources and effects of noise pollution along the frequency.

negative dielectric characteristics and permeability differ from ordinary materials with positive attributes [6]. Metamaterials are divided into thermal, acoustic, electromagnetic, and mechanical classes based on their modified properties, as shown in Figure 2c. Metamaterials are not necessary in nature, but their microstructure is architected or engineered to achieve certain functionalities or several functions (i.e. multifunctional) [7]. These metamaterials must satisfy certain functional requirements, such as a high absorption coefficient over a broad frequency spectrum, cost-effectiveness, structural integrity, lightweight properties, flame retardancy, etc., to be suitable for use in acoustic applications. The discipline of metamaterials is concerned with designing and engineering such materials, with acoustic metamaterials (AC-Meta) being a subset concerned with controlling

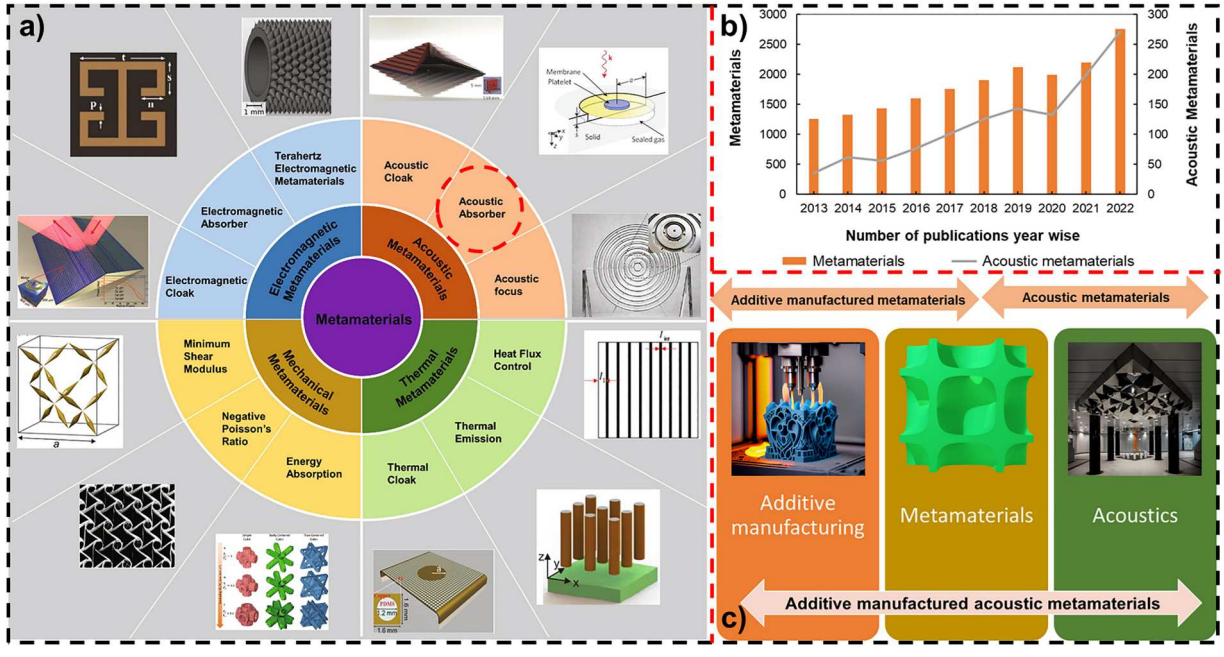



**Figure 2.** Focus of the research: a) applications of the metamaterials with the acoustic application highlighted [7] Reproduced with permission from Materials Today. Rights obtained from Elsevier, b) interest in research topics indicating several publications over the last ten years, and c) selection of topics for the current review.

acoustic waves. AC-Meta are artificial structures comprised of cellular arrays that can alter sound waves on the sub-wavelength scale. These materials have a wide range of effective density and bulk modulus, two extremely important physical characteristics for sound wave propagation. AC Meta have been explored for over two decades, and their complicated features can lead to highly unusual functions [8]. There has been a surge of interest in metamaterials, with researchers particularly concerned with the acoustic applications of metamaterials. It appears from the publications listed on the Scopus website under the topics 'Metamaterials' and 'Acoustic Metamaterials'. There are over 25000 articles with the keyword 'Metamaterials' and over 1500 articles with the keyword 'Acoustic Metamaterials' in the published literature. Acoustic metamaterials exhibit negative density, anisotropic mass properties, and negative elastic modulus. These metamaterials are typically created by combining two or more materials with different mass densities and bulk moduli. Acoustic metamaterials have demonstrated exceptional performance in acoustics, offering remarkable capabilities over a broad range of frequencies. Their unique ability to manipulate sound waves beyond the limits of conventional materials has paved the way for improved noise control and absorption techniques. These materials are engineered to meet specific functional requirements, such as targeting precise frequency ranges, achieving high sound absorption coefficients, and maintaining structural integrity in varied environmental conditions. Additionally, they offer tunability, allowing for customisation in applications requiring precise acoustic tuning. Their versatility and effectiveness across a wide frequency spectrum make them a promising focus for future research and advancements in acoustic technology, particularly in optimising the balance between absorption, reflection, and durability [9].

Figure 2b shows the number of publications published on 'Metamaterials and Acoustic Metamaterials' over the last ten years. This data indicates that the researchers working on acoustics desire to explore the potential applications of AC Meta across numerous dimensions. The broader application of AC-Meta is limited by the substantial fabrication challenges of current technologies, particularly when producing intricate structures. Recently, additive manufacturing (AM) has emerged as a key method for creating 3D metamaterials with complex architectures [10]. Additionally, noise sources span a wide range of frequencies, making them difficult to control using a single absorber. However, acoustic absorbers can be tailored to address specific frequency ranges, which makes customisation a critical aspect of their design and development. Consequently, methodologies for developing acoustic absorbers should offer the flexibility to fine-tune their performance across various frequency spectrums for optimal sound absorption. Additive manufacturing is one such technology that offers numerous options to optimise both process and product parameters, supporting this level of customisation [11].

Therefore, the focus of this review is threefold: additive manufacturing, metamaterials, and acoustics, as shown in Figure 2c. The published review studies are mainly concerned with additively manufactured and acoustic metamaterials. The review papers on additive manufacturing of metamaterials mainly deal with the production of metamaterials, their mechanical strength, and other properties [12,13], whereas those review papers on acoustic metamaterials deal with the acoustic applications of metamaterials as well as the development of acoustic metamaterials using various conventional techniques, and they do not cover the acoustic metamaterials developed by AM in detail [14–19]. Therefore, this review article focuses on additively manufactured acoustic metamaterials, providing a comprehensive overview of their classification and the fabrication techniques made possible through different AM methods. It also covers the evaluation techniques for assessing the performance of these metamaterials and investigates how various process and design parameters impact their sound absorption capabilities. The review emphasises the potential of 3D printing to offer precise control over design features, leading to enhanced acoustic performance for diverse applications.

The review is structured into seven distinct sections. The first section sets the stage with an introduction and background on additively-manufactured acoustic metamaterials. The second section explores various applications of these metamaterials. In the third section, the classification of acoustic-absorbing metamaterials is considered. Section 4 discusses the AM techniques used to create these materials. The fifth section provides an insight into the theories behind acoustic absorption and the methods used to measure it. The sixth section critically examines various studies on acoustic-absorbing metamaterials, highlighting the impact of different parameters on acoustic absorption. Finally, the seventh section concludes the review with a summary and future outlook, proposing research ideas and identifying gaps to guide ongoing research in this field.

## 2. Applications of AC-Meta

Various distinctive acoustic phenomena are produced whenever acoustic waves approach a metamaterial

due to the interaction between the waves and the specially built structures. The applications for such phenomena include acoustic cloaking, acoustic imaging, acoustic wave guiding, acoustic control, and many more, as shown in Figure 3. Acoustic imaging employs an acoustic lens that offers outstanding manageability and concentrating intensity [20], acoustic waveguides are actual structures that steer acoustic waves [21], and acoustic cloaking is a modern technology that utilises noise mitigation or acoustic absorbent materials to construct obstacles invisible to acoustic waves [22]. Transmission or absorption are the two methods used to manage acoustics. The effectiveness of sound insulation is evaluated using acoustic transmission loss, which is defined by the ratio of the incident sound power to the transmitted power [23]. The sound absorption coefficient, which is the ratio of energy absorbed to the total amount of energy that each sound wave originally communicated to the medium, may be used to quantify the capacity of a material for absorbing sound in the air [24]. This complexity implies that a material that absorbs certain frequencies might be less effective than others, posing a significant challenge in designing versatile acoustic absorbers. This review will concentrate solely on the acoustic absorbing metamaterials (ACA-Meta) fabricated by additive manufacturing since many researchers are keen to fully understand the curves of acoustic absorption to potentially apply them to diverse sectors, as illustrated in Figure 4. The review article by Li et al. [25] provides a comprehensive overview of research on additively manufactured structures for acoustic absorption applications, although it primarily emphasises lattice structures. This limitation is noteworthy, as the scope of acoustic metamaterials extends beyond lattice configurations. The present review aims to fill this gap by offering a broader perspective, encompassing various structures classified as metamaterials. It critically examines the influence of different parameters on acoustic absorption, thus providing a more holistic understanding of the field. This approach acknowledges the initial review's contributions and highlights the necessity of exploring a wider range of metamaterial designs for effective sound absorption.

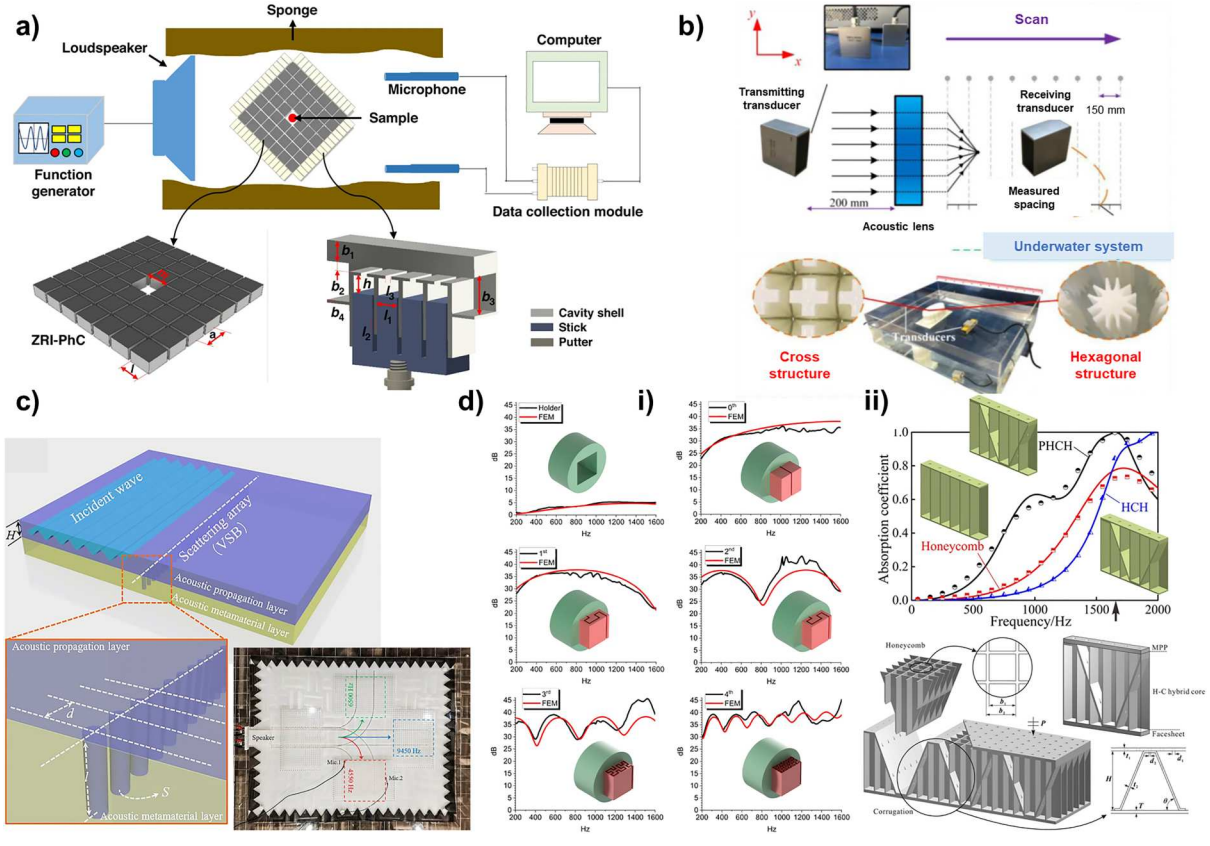



**Figure 3.** Applications of acoustic metamaterials: a) acoustic cloaking [22], Reproduced with permission from Microsystems & Nanoengineering, under the Creative Commons Attribution License (CC BY), b) acoustic imaging, c) acoustic waveguide, and d) acoustic control: i) transmission, ii) absorption [20,21,26,27], Reproduced with permission from Scientific Reports, under the Creative Commons Attribution License (CC BY).

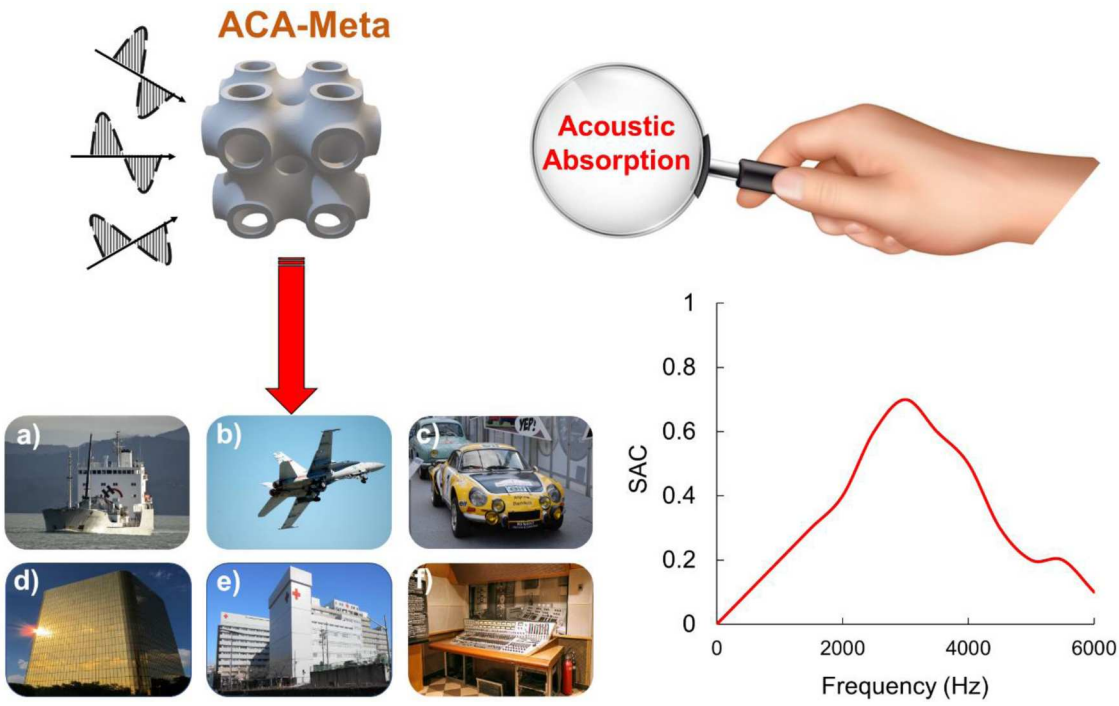



**Figure 4.** ACA-Meta, with its typical acoustic absorption curve, is to be potentially applied at various sectors: a) marine, b) aerospace, c) automotive, d) architectural and construction, e) healthcare, and f) entertainment.

## 3. Classification and sub-classification of ACA-Meta

Acoustic absorbing metamaterials (ACA-Meta) can be broadly categorised into three classes based on their structural characteristics, i.e. perforated, slotted, and cellular, when all the investigations are combined. Structures are designated hybrid ACA-Meta if they share more than two geometries from these categories. The classification of ACA-Meta is shown in Figure 5. Whereas simple perforated, slotted, and cellular configurations are traditionally perceived as distinct from metamaterials, their characterisation changes significantly when developed through AM techniques. AM introduces intricate alterations to a structure, leading to notable enhancements in its performance compared to conventional materials and structures. This transformative effect has prompted the classification of these structures as metamaterials. The incorporation of AM imparts extraordinary properties, influencing specific material characteristics. This recognition challenges the conventional categorisation of these structures, emphasising the impact of advanced manufacturing techniques on their behaviour, thus warranting their designation as metamaterials.

### *3.1. Perforated ACA-Meta*

Perforated ACA-Meta is a panel with many perforated holes in it [19] and was used by researchers to increase the acoustic condition of a building environment. Perforated ACA-Meta is a reclaimable, non-flammable, and ecologically friendly acoustic absorber. It dampens sound through viscous friction in small holes normally spaced apart from a hard surface. When the velocity of the acoustic particle in the pores is at its highest, these absorbers are most efficient [28]. Perforated ACA-Meta provides reasonably good sound absorption capabilities at mid and higher frequencies. However, it has limited sound absorption capabilities at lower frequencies [16]. The diameter of the perforations and thickness of the panel are typically less than 1 mm in the case of the micro-perforated panel (MPP). The overall surface area of the panel is generally between 1% and 2% of the total surface area of the micro-perforated pores. MPP absorbers have been used effectively in various applications, including construction machinery, architectural rooms, and exhaust systems. It is appropriate for an interior wall barrier, windscreen or fluorescent light covering, sunshade, or column cover [15]. To efficiently absorb sound, an air gap must be provided between perforated ACA-Meta and a rigid wall [20]. In many situations, typical MPPs are homogenous; however, a mix of perforations with varying sizes is often employed to improve the diversity of layouts. Perforated panels from first-generation structures were composed of metal and featured circular laser-cut perforations. Second-generation perforated panels are less costly since they are made via etching, cutting, shearing, or

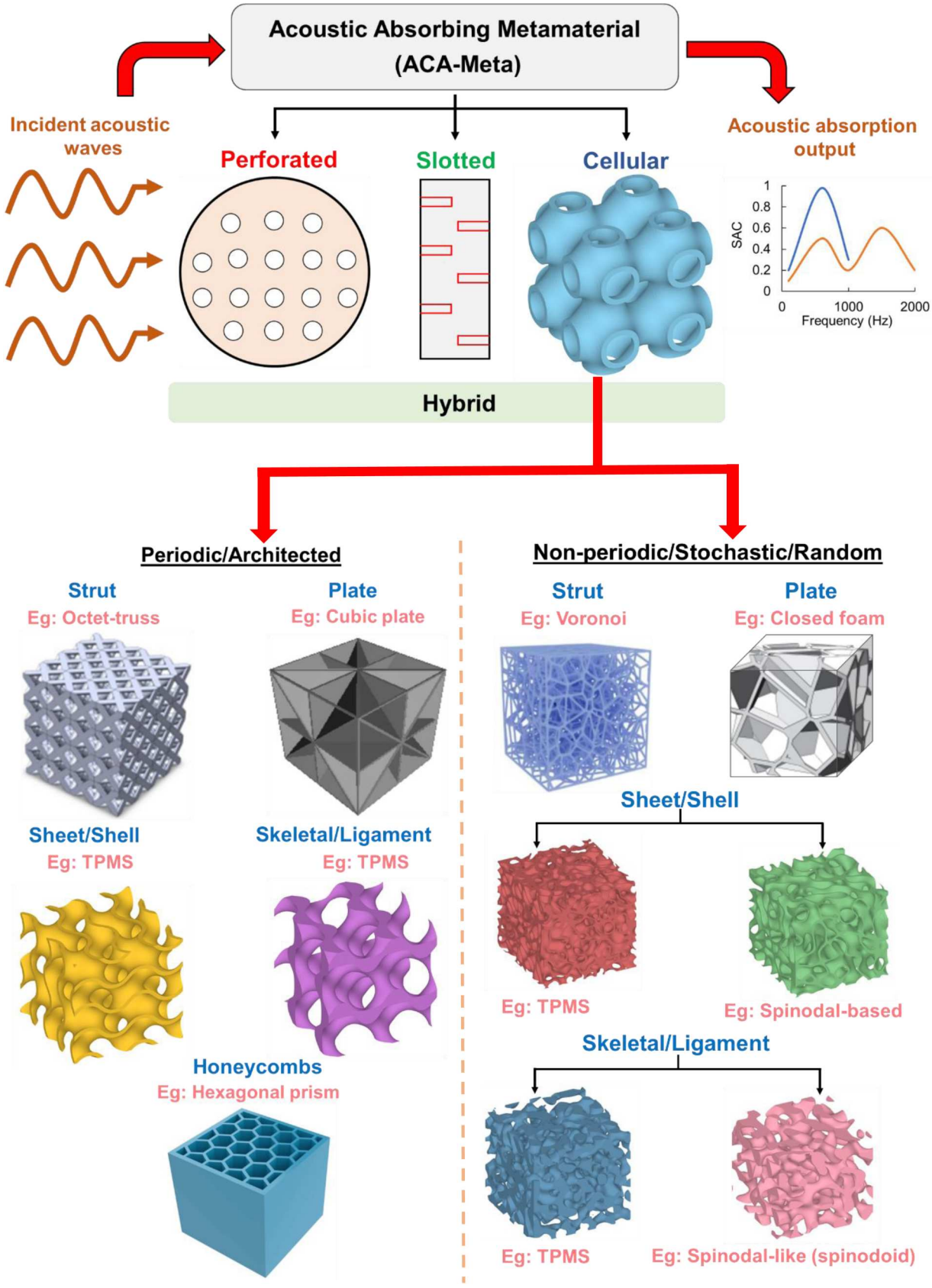



**Figure 5.** Classification and sub classification of ACA-Meta.

grinding [29]. In the current generation, they are fabricated and tuned based on applications using sophisticated technology like AM.

### 3.2. Slotted ACA-Meta

Slotted ACA-Meta are structures with slots instead of circular perforations that work in all practical ways, like the perforated ACA-Meta. This includes structures with resonators and spaced coiled structures (Hilbert, fractal, labyrinthine, spiral, and helix). When a sound wave enters the slotted ACA-Meta, it travels along the slotted path, which is significantly longer than the physical dimensions of the structure, efficiently lowering the rate of sound while decreasing the intensity of sound due to the frictional resistance provided at the air channel wall interface [30]. For slotted absorbers, 3D printing is a flexible technique offering design freedom, customisation, and a fast-manufacturing process. It is typically more resource-efficient than conventional subtractive processes, reducing material waste while providing an improved sustainable manufacturing approach. It is scalable, making it appropriate for either smaller-scale or large-scale manufacturing, and it lowers manufacturing expenses, enabling it affordable to create bespoke slotted absorbers in small numbers [31].

### 3.3. Cellular ACA-Meta

Cellular structures have attracted considerable interest because of their lightweight, high strength, energy absorption, and vibration reduction properties. Gibson and Ashby coined the term 'Cellular Structure' for the first time, believing it comprised foams and honeycombs [32]. These materials display a wide range of topological characteristics, length levels, and structurally controlled properties, such as enhanced absorption of vibration, a significant strength-to-mass ratio, a variety of topological features, length distributions, and structural breakdown regulation [10]. Because of their superior qualities, cellular structures are frequently employed in the aerospace, bioengineering, and robotics sectors and various technological domains. The use of AM technologies has accelerated the development of cellular structures [33]. These cellular structures have also proven to exhibit excellent acoustic properties [34]. Cellular ACA-Meta can be periodic or non-periodic based on periodicity. Periodic cellular ACA-Meta can be further sub-classified into strut, plate, sheet/shell, skeletal/ligament, and honeycombs. In contrast, in the case of non-periodic, spinodal topologies get added without the honeycombs. Periodic strut-based structures comprise a network of prismatic struts joined at vertices, creating a truss-like framework that efficiently distributes load, making them suitable for high strength-to-weight ratio applications [35]. These may include octet-truss, face-centered cube, body-centered cube, Kelvin cells, and other geometries usually connected by struts. On the other hand, non-periodic trusses, such as those based on Voronoi diagrams, employ a seed-based spatial division where points in each cell are closer to the seed that generated them, resulting in cells that form planes with convex polygons without holes in 3D and line segments dividing the space in 2D. These cells maintain the characteristic of being hole-free convex polygons in both dimensions [36]. Periodic plate-based structures, another category, involve plates periodically situated between adjacent trusses, offering a unique combination of properties [37], whereas, in the case of non-periodic plate-based structures, the position of plates does not follow a regular pattern. These include structures like closed-cell foams, quasi-random foams, etc. [38]. TPMS-based structures like gyroid, Neovius, primitive, and diamond may be formed by either thickening the minimum surface to form periodic sheet-based cellular structures or hardening the volumes encompassed by the minimal surfaces to form periodic skeletal-based cellular structures [39]. Non-periodic TPMS are achieved by generating randomness on heterogeneous structures using control points and sub-domains [40]. Further, spinodal structures are non-periodic architectures with bi-continuous cellular layouts that replicate the spinodal topologies that form during the early stages of diffusion-driven phase separation in a homogeneous solution. Spinodal topologies, regulated by the linear Cahn–Hilliard equation, could be modelled using a Gaussian random field created by superposing plane waves with a wavelength that is set but at random sampling wave vectors and phase shifts [41]. These structures are categorised as spinodal-based and spinodoid for sheet and skeletal structures, respectively. Finally, the regular form of honeycomb structures in periodic cellular setups, with identical dimensions and shapes of unit cells, encompasses a variety of cell forms like tetrahedrons, triangular prisms, and hexagonal prisms [42].

### 3.4. Hybrid ACA-Meta

Amid the extensive discussion on perforated, slotted, and cellular acoustic metamaterials, hybrid ACA-Meta appears as a combination, incorporating components from two or more configurations. This combination aims to achieve a synergistic impact by using the many advantages of different structures.

## 4. Additive manufacturing of ACA-Meta

Conventional methods such as casting, molding, machining, and laser cutting have been commonly

used to manufacture acoustic absorbers. Still, they face several limitations, particularly in customisation, controlling design parameters, and achieving high inner porosity [43]. For example, Daniel et al. (2018, 2020) explored compression molding to create PLA-based absorbers reinforced with kenaf and coconut fibres, but noted challenges in tailoring these materials for specific acoustic performance [44,45]. Similarly, Daeipour et al. (2017) and Jayamani et al. (2014) developed absorbers reinforced with natural fibres like wood, rice straw, and kenaf, observing good performance but limited flexibility in tuning their properties. The process often involves complex and time-consuming steps such as fibre preparation, blending, and perforation. Taking these limitations into consideration, AM presents a promising alternative. It offers enhanced design flexibility, customisation, and the potential to overcome many challenges faced in conventional techniques, particularly in developing absorbers with tailored acoustic properties across different frequency ranges [46,47].

Additionally, AM technologies have more regulated parameters and stronger active interaction between material qualities and process variables, allowing for the development and optimisation of complicated designs. This method, also known as 3D printing, combines materials to build three-dimensional things layer by layer and is employed in various production areas, such as automotive, biomedical, and aerospace. Additive manufacturing is divided into multistep and single-step techniques, subdivided into various categories as per ISO/ASTM 59000:2021, as shown in Figure 6 [48,49].

The technique of vat photopolymerization involves treating visible or ultraviolet light with a light-curable resin that is kept in a vat. The polymerisation reaction is sparked by this curing light, which either forms chains of polymers or crosslinks them to produce a solid resin. It is impossible to reverse the process and return it to a liquid state. A sliced STL file is used to produce successive layers of resin. Stereolithography (SLA) and digital light processing (DLP) are two curing techniques used in photopolymerization. Using an SLA 3D printer, a platform is submerged in a clear resin-filled tank, and a laser beam is used to trace the building area and cure it using the STL template. After that, the platform is moved up or down in the z direction, and the curing procedure is repeated layer by layer until a 3D sculpture is finished. In contrast to SLA, DLP technology uses a digital light projector to reflect a laser source rather than a mirror during the curing process. The resolution of the projector affects the accuracy of printed parts [51].

Powder bed fusion (PBF) entails constructing a platform bed covered in small layers of very fine particles. The extra powder is vacuum-extracted after fusing the particles with a liquid binder. There are three methods for preparing surfaces: coating, sintering, and infiltration. The quality of the final element is affected by powder particle size, binder viscosity, and powder spreading speed. Low melting or sintering temperatures can be addressed with lasers. The method provides flexibility in terms of material choice and room temperature requirements. By providing support, the powder bed overcomes obstacles in material removal. On the other hand, it is more expensive, takes longer, and results in more porosity during powder fusion [52]. Direct metal laser sintering (DMLS), selective laser melting (SLM), and selective laser sintering (SLS) are some of the main PBF techniques. Because of the reduced beam size, these technologies create precise and well-finished products for small-scale manufacturing. While SLM melts alloy powder particles to produce a shorter production cycle time and better microstructure at high cooling rates, SLS requires distributing powder over a constructed platform. Aerospace parts are produced more efficiently and economically using DMLS, comparable to the SLM technique, yield functional prototypes, short-run components, and functionally graded materials [53,54]. Material jetting (MJ) technology is an AM technique that produces functional components with reduced surface roughness and great geometric precision using a liquid photopolymer. Its low purchase-to-fly ratio and layer-by-layer manufacturing technique make it attractive in biomedicine, dentistry, manufacturing, and aviation. After being heated in the transmission path, photopolymer ingredients are deposited as droplets on the build platform, and MJ stores them in air-excluding containers. To cure the molten material, ultraviolet light is shone upon it. Because of its benefits, which include adjusting to small layer thicknesses, providing two options for surface texture, and requiring no post-processing, MJ is a well-known AM technique in the polymer printing industry. A single, multi-material approach also permits blending several materials, including PLA, ABS, and polyamide, facilitating the creation of composite components for particular uses. Because of their closed ambient chamber, MJ printers are appropriate for both families and workplaces [55]. The two words are used interchangeably to refer to material jetting methods. These names are used because the material jetting printer manufacturers Stratasys (PolyJet) and 3DSystems (MultiJet) have gained branding licenses [55].

Binder jetting is an AM technology that uses a liquid bonding agent to selectively unite powder particles layer by layer to construct objects from 3D data. It has distinct features such as the capacity to handle

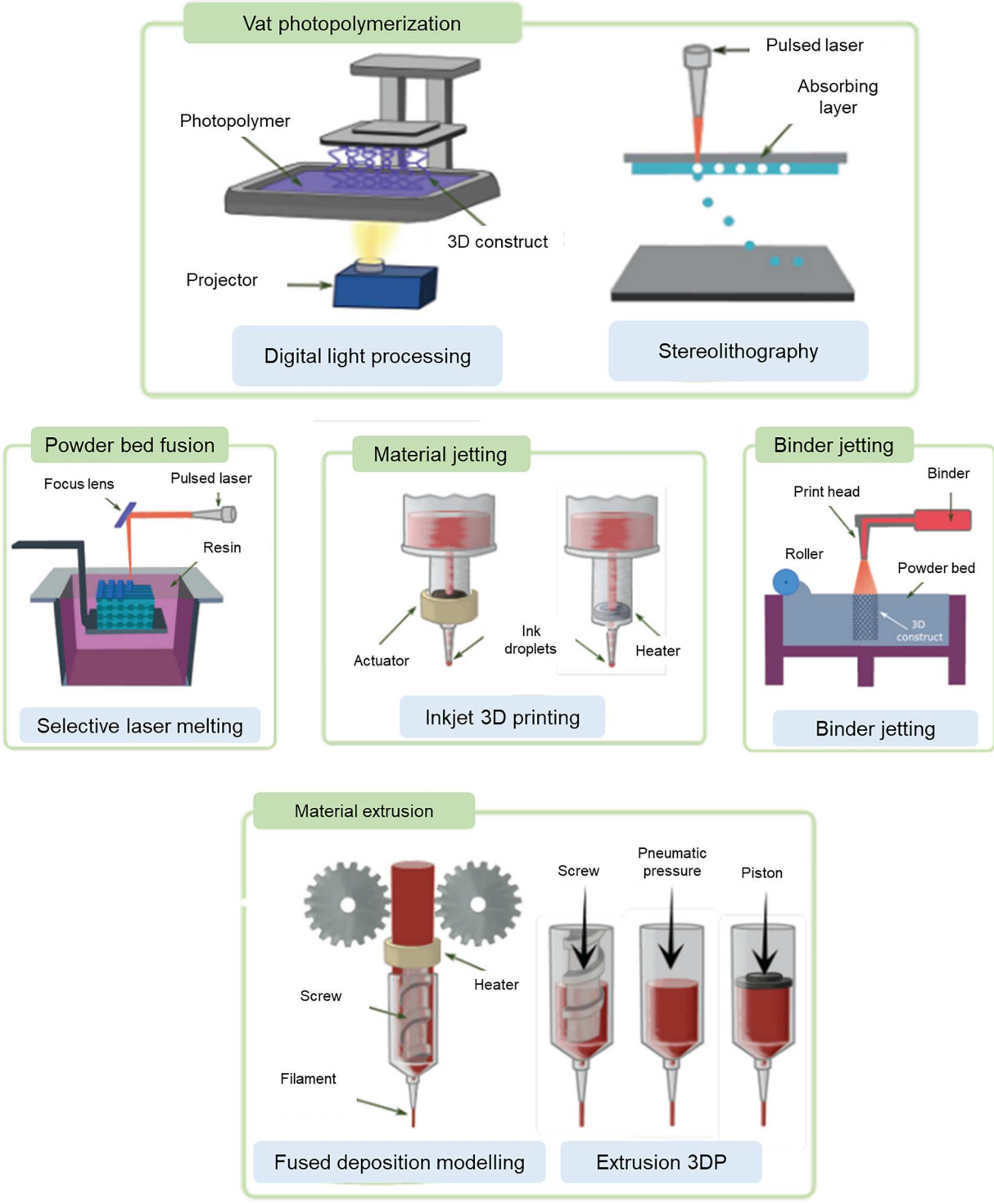



**Figure 6.** Additive manufacturing techniques (ISO/ASTM 59000:2021) [50], Reproduced with permission from International Journal of Bioprinting, under the Creative Commons Attribution License (CC BY).

different materials, the lack of a need for an explicit support structure, and great scalability. The binder is often blasted to the whole cross-section of each layer when printing an item with binder jetting [56]. A print head is used in colour jet printing (CJP) to preferentially disperse the binder into a powder layer [57]. Directed energy deposition (DED) melts feedstock material with a high energy density heat source, generating a metal track and making a 3D near-net-shape component. Commercially accessible methods are known as laser metal deposition (LMD), direct metal deposition (DMD), or laser-engineered net shaping (LENS), depending on

the energy source and the form of the material [58]. Sheet lamination is a 3D manufacturing technique that stacks and laminates thin paper, plastic, or metal sheets. Laser cutting, or CNC machining, produces the final form. For various uses, several processes, such as laminated object manufacturing (LOM), ultrasonic additive manufacturing (UAM), and plastic sheet lamination (PSL), are utilised. LOM, one of the earliest commercially available AM processes, uses paper as a material and adhesive rather than welding. Sheets or rolls of materials, including paper, ceramics, polymeric metal-filled tapes, and composites, are machined and sealed [59]. Material extrusion, popularly known as fused deposition modelling (FDM), deposits a material extruded from a nozzle in each layer in a pattern of parallel lines. However, the mechanical qualities of the finished products are limited since a relatively limited number of materials, including thermoplastics and a few engineering polymers, have been employed as feedstocks for FDM. Consequently, FDM procedures are required to research materials and produce composite products to make this technology appropriate for generating functioning, load-bearing parts [60]. Direct ink writing (DIW) is also an extrusion-based approach employed in this study because liquid-phase ink is poured from the nozzle at controlled flow rates and deposited over digitally set paths to form three-dimensional things layer by layer [61]. Table 1 provides the AM techniques, commercial names, merits, and challenges to be considered for fabricating ACA-Meta.

A few studies have been conducted in a comprehensive exploration of the capabilities of AM for acoustic applications, each contributing unique insights into the fabrication and performance of ACA-Meta. ACA-Meta fabricated by various additive manufactured techniques is shown in Figure 7. A notable study focuses on constructing a frequency-dependent cell transfer matrix using a complete viscothermal acoustic model. It examines the micro-to-macro dynamics within a single-cell structure. Two identical Kelvin cell structures were fabricated and analyzed using the DLP and SLM technologies. These methods were chosen for their precision in depicting complex geometries, focusing on the impact of geometry and surface finish on acoustic properties [63]. Another research area investigates the feasibility of reproducing porous samples designed for sound absorption via additive printing techniques. Different research centres do independent production and acoustic measurements using the same geometric codes that depict agreed-upon periodic cellular patterns. Tests conducted on samples with the same cellular design produced largely consistent results; however, variations in the results might result from surface imperfections caused by the manufacturing process, including microporosity. SLA technology is recommended as the best way to create samples since it reproduces specified periodic shapes well. It is particularly helpful for prototypes, validation, and sophisticated acoustic modelling of porous and meta-porous materials with a planned periodic shape [64]. Superior SLA 3D printers may create materials with a planned periodic network of pores, but these features are more substantial than traditional acoustic foams. Affordable equipment, such as conventional 3D-printing resolutions used in SLA and other AM technologies, cannot enhance these qualities, particularly when 3D printing from powder-based materials, such as laser sintering, laser melting, or binder jet 3D printing processes, the microporosity of the 3D-printed skeleton might happen. The 3D-printed skeleton's sufficient open microporosity results in beneficial double porosity, significantly impacting how sound waves travel through and are absorbed by such materials. The analysis demonstrates that pressure diffusion's potent sound absorption effects outweigh shape – and surface-related uncertainties and flaws [65]. Further, a study presents an in-depth analysis of 3D

**Table 1.** Additive manufacturing techniques, commercial names, merits, and challenges to be considered for fabricating ACA-Meta [62].

| S.No | Additive manufacturing techniques | Commercial names | Merits | Limitations/challenges |
|---|---|---|---|---|
| 1. | Extrusion | Fused filament fabrication (FFF)/fused deposition modelling (FDM), direct ink writing (DIW) | Economic, large variety of materials available | Limited precision |
| 2. | Vat photopolymerization (VPP) | Stereolithography (SLA), digital light processing (DLP) | High precision and surface finish | Material limitation, slow build process |
| 3. | Powder bed fusion (PBF) | Selective laser sintering (SLS), selective laser melting (SLM), direct metal laser sintering (DMLS) | Economic, large variety of materials available | Slow process, high power required |
| 4. | Material jetting (MJ) | PolyJet (PJ), multijet printing (MJP) | High precision and low waste | Material limitation |
| 5. | Binder jetting (BJ) | Colour jet printing (CJP) | Economic and high speed | Concern on mechanical properties |
| 6. | Direct energy deposition (DED) | Laser-engineered net shaping (LENS), direct metal deposition (DMD), | High quality of prints | Material limitation |
| 7. | Sheet lamination | Laminated object manufacturing (LOM) | High speed and low cost | Material limitation |

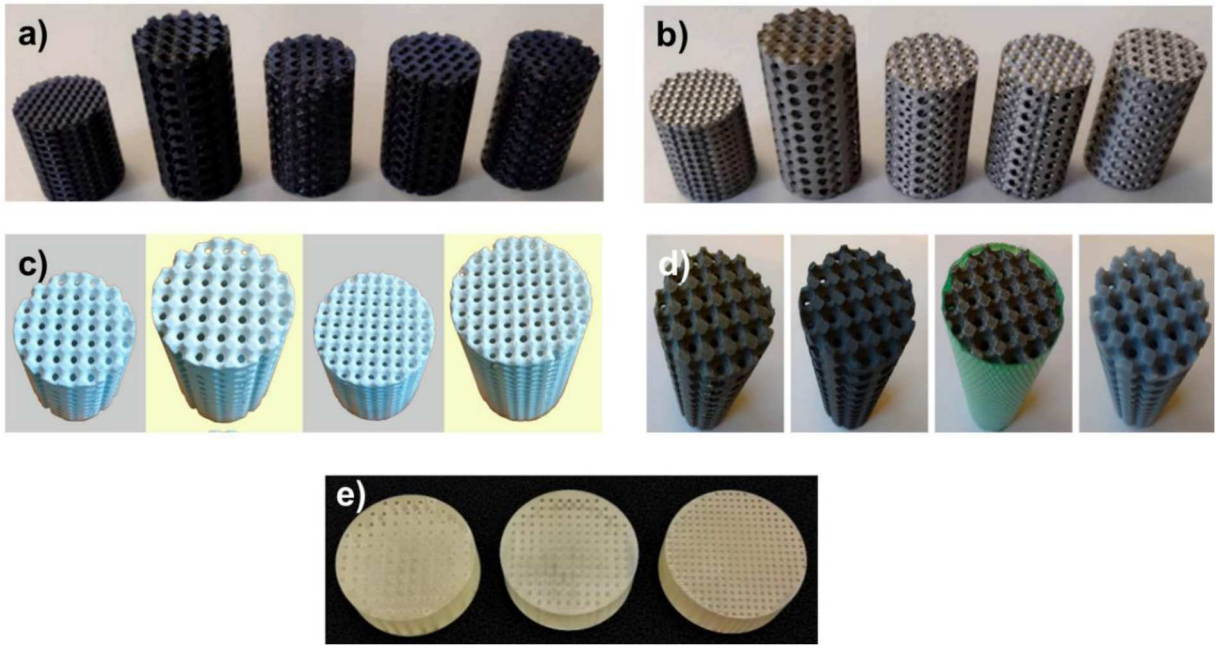



**Figure 7.** ACA-Meta fabricated by various additive manufactured techniques: a) vat photopolymerization, b) powder bed fusion, c) binder jetting, d) extrusion, and e) material jetting [65], Reproduced with permission from Applied Acoustics, under the Creative Commons Attribution License (CC BY), [64] Reproduced with permission from Additive Manufacturing. Rights obtained from Elsevier, [69] Reproduced with permission from Australian Acoustical Society, under the Creative Commons Attribution License (CC BY).

printing of structured acoustic absorbent materials fabricated by DLP and SLA at microscopic scales. The design idea for an acoustic benchmark metamaterial is presented in this research. It consists of a periodic network of interconnected resonant chambers with cubes with spherical interior cavities connected by cylindrical holes. Variations in surface roughness and dimensional fidelity affect the values of the coefficient of absorption that are determined experimentally. One major obstacle to employing DLP and SLA polymer resins to produce finer-scale pieces is their decreased viscosity [66]. Additionally, an investigation has been performed to compare the acoustic properties of two extrusion-based 3D printing technologies: extrude-and-pull and fibre bridging. The extrude-and-pull technique yields thin, hair-like fibres, but the fibre-bridging approach creates thicker, more sound-absorbing fibres. The extrude-and-pull approach is a simple addition to processes already used in AM to add fibres to a base surface. Extrude-and-pull samples have reduced absorption performance over the frequency range under investigation due to their reduced individual fibre thickness, which also causes reduced visco-inertial and thermal losses [67]. Lastly, aerogel-based acoustic absorbers with regulated microstructures were successfully printed using a novel 3D printing technique that combines extrusion and freeze-casting. The bulk absorbers of cellulose nanocrystal aerogel exhibited wide and significant acoustic absorption qualities, attracting attention to developing extremely lightweight, sound-absorbing devices for use in aircraft [68]. These studies provide a nuanced understanding of how different AM technologies can be optimised to fabricate ACA-Meta. They emphasise the importance of selecting the right manufacturing process based on the desired structural complexity, scale, and acoustic performance. Furthermore, they highlight the need to balance the trade-offs between manufacturing precision, material properties, and cost-efficiency in pursuing effective acoustic-absorbing metamaterials.

## 5. Acoustic absorption and its measuring techniques

According to energy conservation laws, the total energy in the incident sound is the sum of absorbed, reflected, and transmitted energy. The absorbed sound, measured in terms of the sound absorption coefficient (SAC), can be mathematically as the ratio of sound energy absorbed by a given surface to sound energy incident on the surface. It ranges between 0 and 1. A sound absorption coefficient of 1 indicates complete absorption, while a coefficient of 0 indicates no absorption. Three main mechanisms contribute to sound absorption: heat dissipation, physical vibration of the material or structure, and frictional dissipation at the viscous boundary layer. When substantial frictional forces are applied to air molecules, they become less mobile and form the viscous

boundary layer. This loss peaks during resonance and plays a major role in sound absorption. Heat transfer across the thermal boundary layer causes thermal dissipation, which causes fluid molecules to oscillate and create frictional forces between neighbouring fluid layers. Viscous losses outweigh thermal losses in terms of contribution. Physical vibration creates frictional forces and physically disperses air molecules, which helps absorb sound, especially in materials with high aspect ratios or flexible shapes [70]. SAC ($\alpha$) can be calculated based on Equation (1) as follows:

$$\alpha = 1 - |R|^2 \tag{1}$$

where the reflection coefficient ($R$) is the ratio of reflected pressure to incident pressure, and the specific acoustic impedance ($Z$) is the resistance provided by the geometries of the acoustic-absorbing medium to the acoustic flow caused by the sound pressure exerted on it. This is similar to electrical impedance, which refers to the resistance of an electrical current passing through an applied voltage. Similarly, $Z$ is a complex number comprising two parts: the real component, which represents acoustic resistance, and the imaginary element, which represents mass reactance. The real portion represents the quantity of energy lost in the system, whereas the imaginary part describes the frequency at which resonance occurs. The following relations in Equation (2) correlate specific impedance ($Z$) to the reflection coefficient ($R$):

$$R = \frac{Z - Z_0}{Z + Z_0} \tag{2}$$

where, $Z_{0\ \text{is}}$ the acoustic impedance of air at standard conditions, calculated by multiplying the speed of sound ($c_0$), by the density of air ($\rho_0$) at normal conditions.

The sound absorption coefficient can be measured numerically, analytically, and experimentally. The numerical simulation known as finite element analysis (FEA) is used to estimate the SAC of these ACA-Meta structures. Unlike analytical analysis, FEA does not require geometry discretization but simulates the full model. The physics of viscous boundary losses and thermal losses are used in FEA methodologies. COMSOL Multiphysics is used for FEA, including acoustics modules simulating the air phase and suitable boundary conditions. Pressure acoustics and thermoviscous modules are applied to impedance tubes and specimens. Thermoviscous acoustics represents the structure for detailed thermal and viscous boundary losses, assuming isentropic acoustics and a lossless system. In contrast, the pressure acoustics module creates incoming acoustic waves and collects pressure data points. Equations (2) and (1) are used to determine sound absorption coefficients, which begin with sound pressures at integration sites on the FEA model [25].

A few notable analytical models for perforated, slotted, and cellular materials are included below. Maa developed a model for predicting the sound absorption of perforated panels based solely on the micro-perforated hole acting as a thin circular tube. Suppose the wavelength of the incident sound wave is more significant than both the distance between the adjacent tubes ($b$) and the perforation diameter ($d$). In that case, the acoustic impedance of the tube ($Z$) can be formulated as mentioned in Equation (3) [71].

$$Z = \frac{j\omega\rho t}{1 - \left(\frac{J_1}{J_0}\right)\left(2r\frac{\sqrt{-j}}{\sqrt{-j}}\left(\frac{\eta}{\rho\omega}\right)\right)\left(\frac{\rho\omega}{r\eta\sqrt{-j}}\right)} \tag{3}$$

$$\omega = 2\pi f \tag{4}$$

where $\omega$, $\rho$, $t$, $r$, and $\eta$ are defined as the sound angular frequency that can be calculated using Equation (4), the density of air, tube length (MPP thickness), tube radius (micro-perforated hole radius), and air dynamic viscosity, respectively. While $J_0$ and $J_1$ are the zeroth and first-order Bessel functions of the first kind, respectively. Equation (3) is divided by the ratio of micro-perforated area to calculate the acoustic impedance of MPP. However, Equation (3) needed a complicated computation and calculating procedure, regarded as challenging and unworkable. Thus, Maa proposed a superior equation with a marginal error of less than 5%, and the normalised acoustic impedance of MPP ($Z$) may be expressed as given in Equation (5) [72].

$$Z = R_n + jX_n \tag{5}$$

$$R_n = \left(\frac{dx\sqrt{2}}{8t} + \sqrt{\frac{x^2}{32} + 1}\right) \times \frac{B_1 t}{pd^2} \tag{6}$$

$$X_n = \left(\frac{1}{\sqrt{\frac{x^2}{2} + 9}} + \frac{0.85\ d}{t} + 1\right) \times \frac{ft}{p} \times 0.0185 \tag{7}$$

$$x = B_2 d\ \sqrt{f} \times\ 10^{-3} \tag{8}$$

where $X_n$ and $R_n$ are defined as the reactance and resistance of MPP, respectively. $f$, $p$ and $d$ are defined as the frequency of incident acoustic waves, a ratio of micro-perforated area, and the diameter of a micro-perforated hole, respectively. $B_1$ and $B_2$ are constants for the normalised equation, and their values for metals are 0.335, 0.210, and 0.147, 0.316 for non-metals, respectively. Finally, the sound absorption coefficient of MPP ($\alpha$) under normal incidence conditions backed by an air-

gap can be expressed as shown in Equation (9) [73].

$$\alpha = \frac{4R_n}{(R_n+1)^2 + \left(X_n\omega - cot\left(\frac{\omega D}{c}\right)\right)^2} \tag{9}$$

where $c$ is defined as the speed of incident acoustic waves. After considering the overperforation effects, the input acoustic impedance for the perforated panel could be measured as provided in Equation (10) and for the slotted absorber as provided in Equation (11).

$$Z = \frac{\sqrt{2\rho_0\omega\eta}}{2\phi} + i\frac{\omega\rho_0}{\phi}\left\{0.85\frac{d}{F(\varepsilon)} + t\left[1 - \frac{2}{s\sqrt{-j}}\frac{J_1(s\sqrt{-i})}{J_0(s\sqrt{-i})}\right]\right\}^{-1} - iZ_0\cot(kD) \tag{10}$$

$$Z = \frac{12\eta t}{\phi d^2}\left(\sqrt{1+\frac{s^2}{18}} + \frac{\sqrt{2}sd}{12t}\right) + \frac{i\rho_0\omega t}{\phi}\left[1 + \frac{1}{\sqrt{25+2y^2}} + \frac{F(\varepsilon)d}{2t}\right] - iZ_0\cot(kD) \tag{11}$$

where $s$ is the area of the edge effects, and $F(\varepsilon)$ is the Fok function. (11)

Delany-Bazley developed an empirical formula for homogeneous and isotropic materials and normalised them into a dimensionless group to determine the characteristic impedance ($Z_c$) and wave propagation constant ($k$) [74]. Because this approach is based solely on the flow resistivity characteristic of the material, their model is regarded as a basic and straightforward model for quick approximation. The empirical relationship between two frequency-dependent factors, characteristic impedance ($Z_c$) and propagation constant ($k$), may be stated in terms of flow resistivity ($\sigma$) as follows [75]:

$$Z_c = \rho_0 c_0[1 + 0.057b^{-0.754} - i(0.087b^{-0.732})] \tag{12}$$

$$k = \frac{2\pi f}{c_0}[0.189\, b^{-0.595} + i(1 + 0.0978b^{-0.7})] \tag{13}$$

In terms of Equations (12) and (13) the surface acoustic impedance ($Z$) of the porous layer while backed with a rigid wall can be calculated as seen in Equation (14):

$$Z = Z_C \cot h\,(k.d) \tag{14}$$

Where $\rho_0$ = air density; $c_0$ = speed of sound in air; $f$ = sound wave frequency; $d$ = thickness of cellular structure; $b = \frac{\rho_{0f}}{\sigma}$.

The model is only relevant for $0.01 \le b < 1.0$. This model may be utilised for a broad range of frequencies and the accessibility of the flow resistivity of the material. However, the actualisation of the model is constrained to a specific range of flow resistivity ($\sigma$), which is $1000 \le \sigma \le 50000$ MKS rayl m$^{-1}$ and porosity close to 1.

Johnson-Champoux-Allard Model is a rigid frame model whereby the solid phase remains constant. Allard and Champoux [76] built on their work [77] to create this model. This model incorporates five non-acoustical parameters: flow-resistivity, tortuosity, porosity, thermal characteristic length, and viscous characteristic length. The following two variables are related to thermal and viscous losses, respectively. The initial expressions for density and bulk modulus are given in Equations (15) and (16):

$$\rho = \frac{\rho_0}{\emptyset}[\alpha_\infty] + \frac{\sigma\emptyset}{j\omega\rho_0}\sqrt{1 + j\frac{4\alpha_\infty^2\eta\rho_0\omega}{\sigma^2\,\Lambda^2\,\emptyset^2}} \tag{15}$$

$$K = \frac{\gamma P_0/\emptyset}{\gamma - (\gamma-1)\left[1 - j\frac{8k}{\Lambda' 2C_p\rho_0\omega}\sqrt{1 + j\frac{\Lambda' 2C_p\rho_0\omega}{16k}}\right]} \tag{16}$$

where $l$ = total fibre length per unit volume; $r$ = cross sectional fibre radius; $\Lambda$ is the viscous characteristic length. $\omega$ = angular frequency; $\gamma$ = ratio of specific heat at constant pressure to specific heat at constant volume; $\Lambda'$ = thermal characteristic length; $\emptyset$ is the porosity, $P_0$ = atmospheric pressure; $\rho_o$ = air density.

The expressions for characteristic impedance, complex wave number, and acoustic impedance may be derived by the following Equations (17), (18), and (19), respectively:

$$Z_c(\omega) = \frac{1}{\emptyset}\sqrt{\rho\,(\omega).\,K\,(\omega)} \tag{17}$$

$$k_c(\omega) = \omega\sqrt{\frac{\rho\,(\omega)}{K\,(\omega)}} \tag{18}$$

$$Z = Z_c(\omega).\cot h(k_c(\omega)) \tag{19}$$

The sound absorption coefficient can be determined by entering the acquired acoustic impedance into Equations (2) and (1), respectively.

There are various experimental methods for calculating the acoustic absorption of a given acoustic specimen, such as ultrasonic methods, free field methods, and interferometric methods. However, the standing wave tube method and the reverberation method are the most widely used methods nowadays since they both provide a well-defined testing environment [78]. The standing wave tube method and reverberation method are also known as the calculation of acoustic absorption by normal incidence and random incidence, respectively, because acoustic waves incident on the

surface are of normal incidence. In contrast, acoustic waves incident on the latter of the sample are of random incidence, as shown in Figure 8 [79].

The reverberation chamber is a specially designed room for acoustic testing, as shown in Figure 8a. The acoustic wave from different directions collides with the surface of the material to be tested at different frequencies but with the same probability. The acoustic absorption is calculated with respect to the change in reverberation time. The reverberation chamber must meet the conditions that follow. The outside surfaces of the chamber all efficiently reflect sound waves, and sound waves reflected from all angles are almost the same, ensuring that there won't be any noticeable changes in the sound pressure within the chamber other than at the area nearest the sound generator. The reverberation chamber capacity must be more than 200 times the size of the noise source volume to estimate the acoustic power from the noise source. Unparallel disparate walls are employed and polished to reflect as much of the sound waves as possible [82]. ASTM C423, the standard test method for sound absorption and sound absorption coefficients by the reverberation room method, and ISO 354, measurement of sound absorption in a reverberation room, are the standards used for calculating the acoustic absorption of the specimen using the reverberation method. They are the commonly used standards for measuring SAC using the reverberation room method.

The standing wave tube, as shown in Figure 8b, measures the sound pressure and particle velocity within a standing wave inside a tube to compute the acoustic absorption of the sample, which is made of bulk materials. To calculate acoustic absorption for a specimen with smaller sizes, the impedance tube method is more suitable. The best method for measuring sound levels depends on a trade-off between convenience and accuracy. Two commonly used methods are the two-microphone method and the single-microphone method. The single-microphone method may be appropriate when only one microphone can be used, while the two-microphone method is preferred for higher accuracy. A noise source, two microphones, an impedance tube, a plunger, a signal analyzer, and an amplifier comprise the setup. The noise signal the impedance tube needs is generated by the noise source and delivered to the tube. The A and B microphone signals are received, which assists in the computation of incident and reflected sound wave pressure values. The sound absorption coefficient is calculated

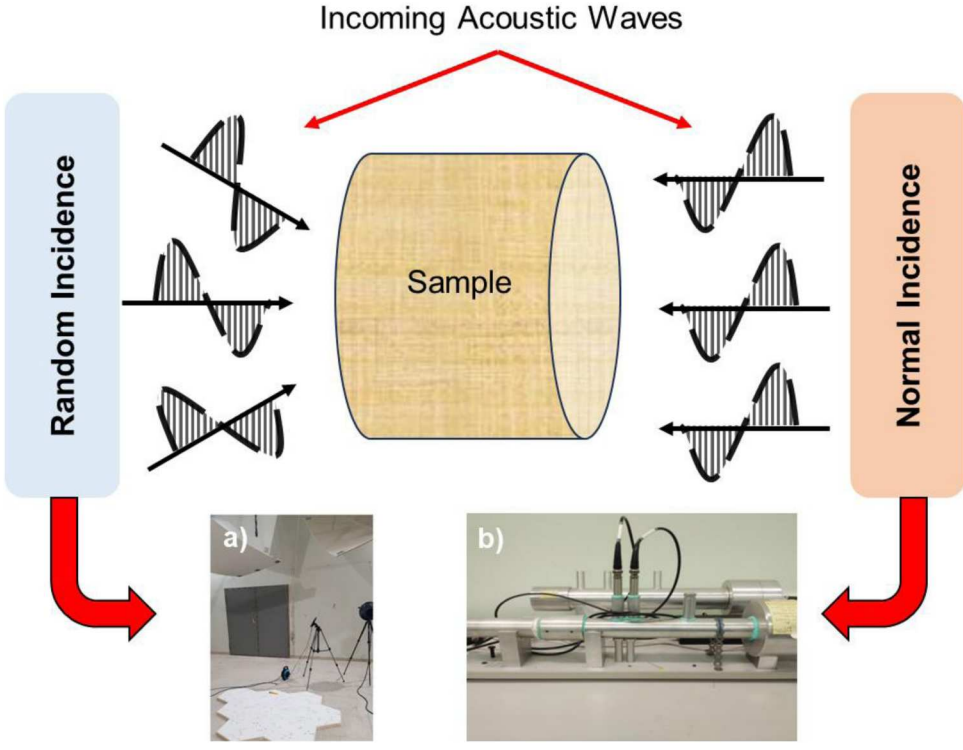



**Figure 8.** Representation of experimental techniques for measuring acoustic absorption: a) reverberation room method [80] Reproduced with permission from Architecture, Structures, and Construction, under the Creative Commons Attribution License (CC BY), and b) standing wave tube method [81] Reproduced with permission from Sound & Vibration, under the Creative Commons Attribution License (CC BY).

by comparing the pressure values. The two-microphone impedance approach depends on monitoring the sound pressure in the impedance tube at two different locations and calculating the acoustic pressure field transfer function among them [83].

The impedance tube method, like the reverberation method, must meet certain requirements, such as the length of the impedance tube being greater than half the wavelength to form a node and wave loop for standing waves. The cross-section area of the impedance tube being less than the length of the sound wave to form plane waves in the tube. As a result, the lowest $(f_{min})$ and highest $(f_{max})$ measurable frequencies for the impedance tube are as follows:

$$f_{max=} \frac{0.6\ \upsilon_0}{D} \tag{20}$$

$$f_{min=} \frac{0.5\ \upsilon_0}{L} \tag{21}$$

where $D$ is the diameter of the impedance tube, $L$ is the length of the impedance tube, and $\upsilon_0$ is the speed of sound in air. The ASTM E1050 standard test method for impedance and absorption of acoustical materials using a tube, two microphones, a digital frequency analysis system, and ISO 10534, the determination of sound absorption coefficient and impedance in impedance tubes using standing wave ratio, transfer function method are the commonly used standards for measuring SAC in the impedance tube.

The following studies address the difficulties of evaluating structures' sound absorption capabilities within an impedance tube. The initial study highlights the resilience of the measurements in different orientations, implying that some geometrical commonalities between the 35 and 60 mm samples are responsible for the consistent outcomes. This measurement stability suggests that the representativeness criteria are satisfied within the given geometric bounds [84]. However, by examining the anisotropic material properties of AM, the second research presents a fresh viewpoint. It indicates the possibility of orthogonality, asymmetric vibrations, and unusual energy absorption behaviours leading to rotationally angle-dependent acoustic absorption coefficients [85]. The two investigations offer divergent conclusions on how sample orientation affects the assessment of sound absorption capacities in an impedance tube. In contrast to one research study, which shows a high degree of agreement across all sample sizes and stresses the insensitivity of measurement repeatability to sample orientation, the other presents the idea of rotational angle dependency, which makes it possible to develop an adaptive sound absorption system. These opposing patterns reveal a more complex picture of how sample orientation affects acoustic absorption.

## 6. Experimental studies

The approach to assessing the acoustic absorption properties of additive-manufactured ACA-Meta involves representing experimental findings through curve graphs with frequency and SAC on the horizontal and vertical axes, respectively. The typical curves exhibit narrow peaks with high intensity, reminiscent of perforated panels and resonators or medium-intensity gradients seen in fabrics, foams, and fibres. However, the complexity of additively manufactured ACA-Meta structures, especially those developed through advanced techniques like AM, results in occasionally combinational and diverse curves. The variances in perforations and cavities (slots) in the structures described by Li et al. [25], as illustrated in Figure 9a, contribute to the variety of resonant peaks.

Compiling these varied trends posed challenges due to their intricate nature. Therefore, a method was proposed to depict trends in ACA-Meta concerning resonant frequency and SAC. Figure 9bi shows two types of structures, 1 and 2, with their respective acoustic absorption peaks along the frequency axis. In Figure 9bii, resonant frequencies, and corresponding SAC values are averaged, with error bars included in cases with multiple peaks. In contrast, direct values are presented for structures exhibiting a single resonant peak. In general, the higher and broader the curve, the better the sound absorption of the material. In certain cases, having more narrow peaks at numerous frequencies is preferred to use it for a wide variety of applications. As a result, in this case, good acoustic absorbers have a single higher and wider peak of acoustic absorption or a resonant frequency with a higher standard deviation (having multiple peaks across the frequency range) and a higher SAC with a smaller standard deviation. This methodology captures the distinctions between additively manufactured ACA-Meta and provides a critical means of interpretation for their complex acoustic absorption behaviour. Utilising the previously outlined methodology, a comprehensive examination was conducted to assess the impact of various structures with different parameters, focusing on the design parameters illustrated in Figure 10. The analysis investigates into the influence of these parameters on acoustic absorption across the frequency spectrum, offering insights into how they affect both resonant frequency and SAC. This thorough review provides an understanding of the intricate relationships

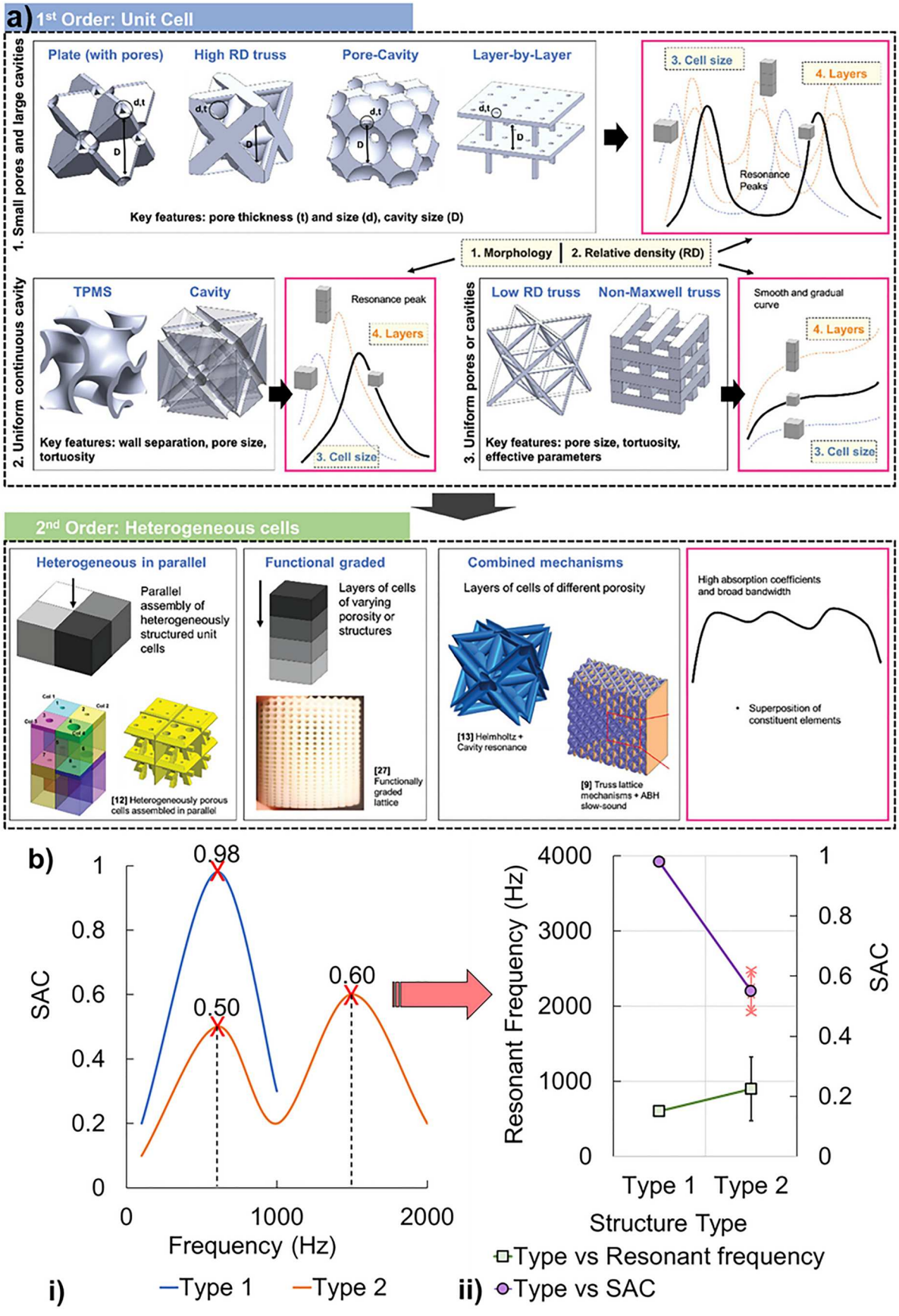



**Figure 9.** Resonant peaks exhibited by ACA-meta: a) different types of peaks by cellular ACA-meta [25] Reproduced with permission from Advanced Science, under the Creative Commons Attribution License (CC BY), and b) methodology to normalise the resonant peaks.

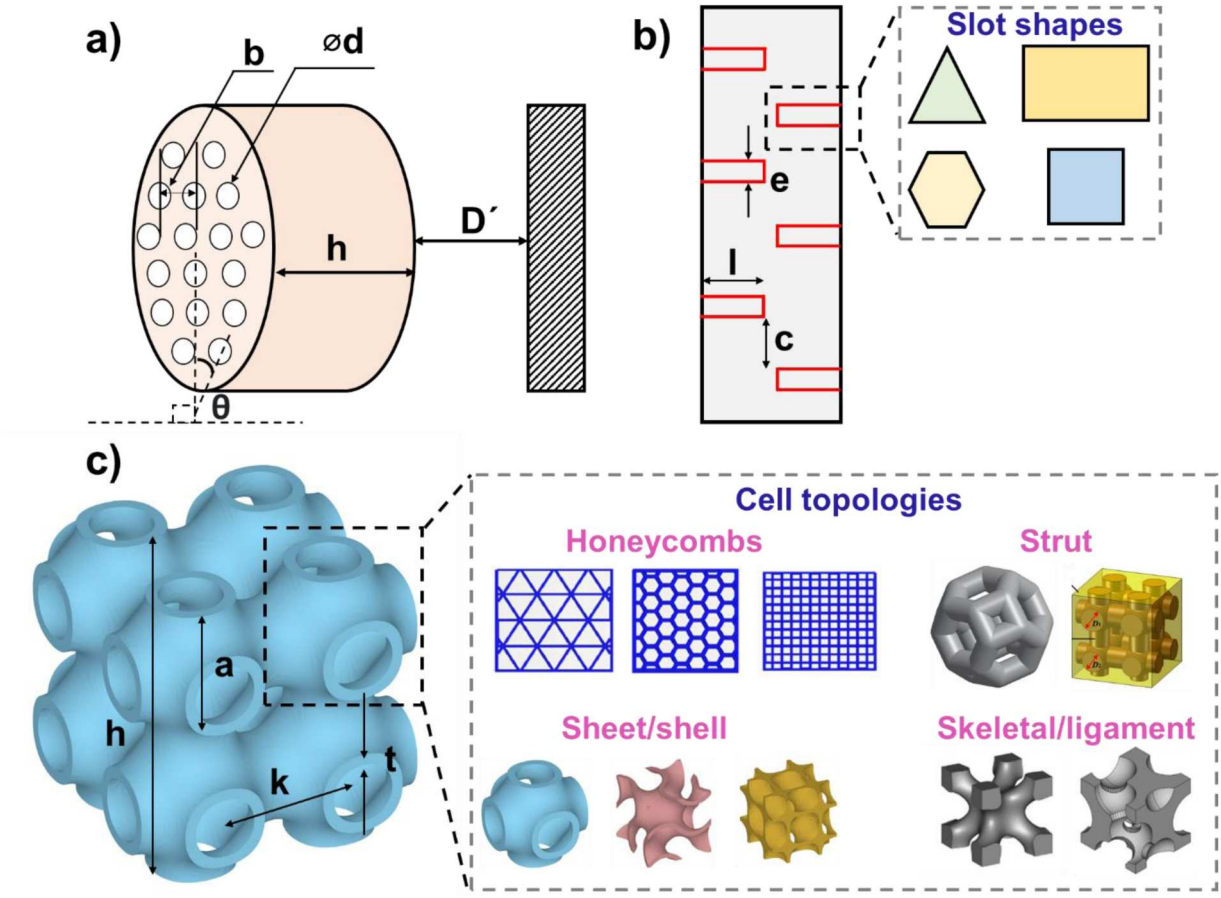



**Figure 10.** Schematic representation and design parameters of ACA-Meta: a) perforated ACA-Meta with the sample height (h), perforation diameter (d), distance between the perforations (b), perforation angle (θ) separated by an air-gap (D´) to the rigid wall, b) slotted ACA-Meta with the slot length (l), slot thickness (e), slot spacing (c), and c) cellular ACA-Meta with the sample height (h), cell size (a), cell spacing (k), shell thickness (t).

between design parameters and the acoustic performance of the material.

### *6.1. Studies on additive-manufactured perforated ACA-Meta*

The field of acoustic material research, particularly with the integration of AM technologies such as 3D printing, is witnessing significant strides in enhancing the sound absorption capabilities of various materials. A deeper exploration of this domain reveals insights into how design and material choices impact acoustic performance. The acoustic absorption characteristics of an additively manufactured perforated ACA-Meta, as shown in Figure 11, were studied. The perforated layers are printed using polymer material with different hole spacings to provide different perforation ratios. The frequency of the maximum sound absorption coefficient can be altered by adjusting the perforation ratio. The results show that increasing the perforation ratio increases the acoustic resonance frequency of the peak sound absorption coefficient, as seen in Figure 12d. The perforation ratio must be adjusted to obtain the desired peak acoustic absorption frequency and a wider frequency range [86]. Concurrently, this work probes into the influence of perforation volume, comprising both the thickness and the diameter of perforations, which change their sound absorption. The acoustic absorption of the perforated ACA-Meta is influenced by resistance and reactance, with the perforated hole playing the most important role. The early peaks of acoustic absorption were minimised as the perforation volume increased [87]. Further research demonstrates the consequence of density in defining the SAC of perforated ACA-Meta. The study indicated that decreasing the infill density causes the mass of the MPP to drop, which slightly impacts its resonant frequency in the mid-frequency spectrum. A drop in infill density is generally accompanied by a fall in the density of the 3D-printed samples. Compared to the specimen with a 20% infill density, the MPP with a 100% infill density was denser and more substantial. It

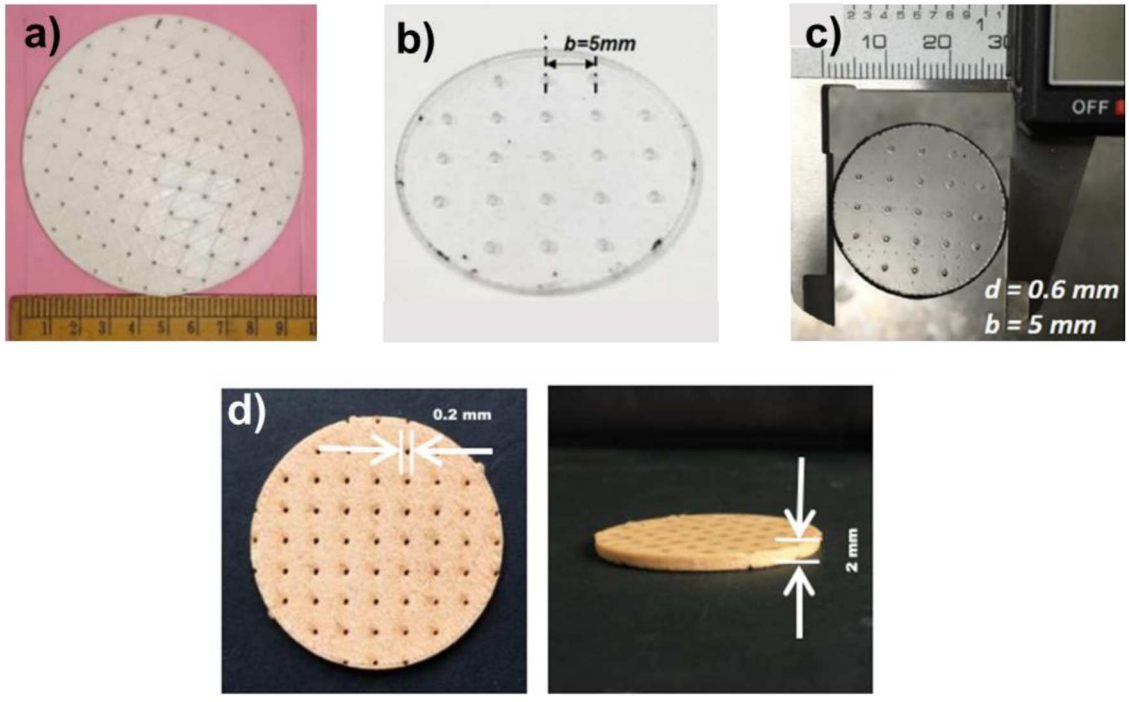



**Figure 11.** ACA-Meta with varying perforation dimensions: a) t = 10.0 mm; d = 1.0 mm [89] Reproduced with permission from Applied Acoustics. Rights obtained from Elsevier, b) t = 1.0 mm; d = 0.8 mm [90] Reproduced with permission from Measurements. Rights obtained from Elsevier, c) t = 1.0 mm; d = 0.6 mm [86], d) t = 2.0 mm; d = 0.2 mm [87] Reproduced with permission from IOP Publishing Ltd, under the Creative Commons Attribution License (CC BY).

was shown that it was possible to decrease the mass while keeping the maximum SAC and without changing the resonant frequency to another spectrum [88].

Research focusing on the acoustic properties of perforated ACA-Meta on geometrical aspects such as porosity, hole diameter, and sample height. Intriguingly, while the diameter of the holes doesn't markedly affect the sound absorption characteristics, as seen in Figure 12b, the sample height does, as seen in Figure 12a, indicating a more complex interaction between physical dimensions and acoustic behaviour [69]. The study of perforated polycarbonate materials further illustrates this complexity. Alterations in the perforation angle decreased sound absorption, as seen in Figure 12c. Yet, strategic adjustments to these angles, coupled with modifications to the air gap behind the sample, can significantly improve absorption, especially at lower frequencies where traditional materials might falter. This suggests that careful consideration of geometric alterations can unlock enhanced acoustic performance in materials traditionally not known for their sound-absorbing properties [91].

Exploring biodegradable materials, research involving polylactic acid indicates that various cross-sectional perforations, such as convergent-diverging, yield superior sound absorption compared to uniform cross-sections. This finding points to the crucial role of internal geometric structuring in optimising acoustic absorption [89]. Further, when 3D printed into a perforated ACA-Meta, investigations into titanium alloys reveal an understanding of material-specific acoustic properties. The studies show that these materials exhibit peak sound absorption coefficients at specific frequencies, enhanced further by aesthetic perforations, namely maze, star, hexagonal, and incorporating a porous back layer. This demonstrates that material composition and structural design are key determinants of acoustic performance [92].

The research explored the efficacy of multi-layered perforated ACA-Meta in enhancing sound absorption, particularly in constrained spaces and at low frequencies. The first study examined a structure comprising four parallel-arranged perforated panel absorbers. With larger apertures compared to traditional perforated panels, these panels demonstrated their ability to absorb sound effectively over a wide bandwidth, particularly in the lower frequency range (120–450 Hz). This design innovation lies in its ability to provide significant sound absorption in constrained spaces using a parallel arrangement of panels [93]. The second study, which probed deeper into the comparison of different layers of perforated ACA-Meta, particularly focusing on their performance across a broader frequency range (200–1600 Hz), revealed that increasing the number of

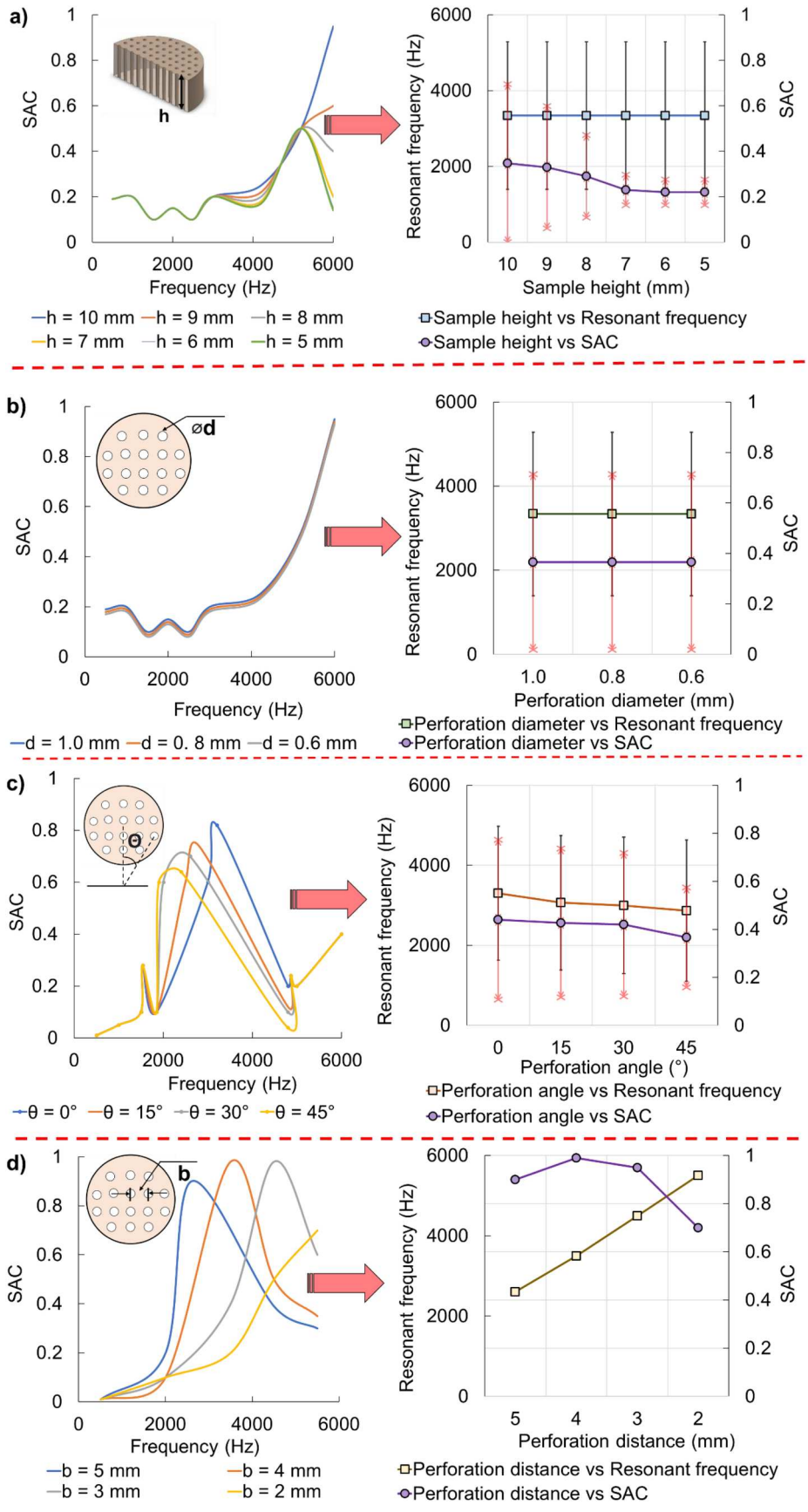



**Figure 12.** Effect of design parameters of the perforated ACA-Meta: a) sample height [69], b) perforation diameter [69], c) perforation angle [91], and d) perforation distance [86] on SAC along the frequency (left) and resonant frequency, SAC (right).

layers not only maintains high absorption coefficients (above 0.8) but also broadens the frequency bandwidth. This is evidenced by multiple peaks on the absorption coefficient curves (two peaks in double-layer and three in triple-layer structures) [94]. In all research studies that investigated the effect of an air gap on SAC, the peak of acoustic absorption shifts towards the low-frequency spectrum as the air gap thickness grows. The perforated end is an acoustic mass, while the air within the impedance tube in this location functions as an acoustic spring. Both correspond to the mass-spring connection and as the thickness of the air gap increases, the stiffness between them decreases, shifting the acoustic absorption maxima to the low-frequency spectrum. When the sound frequency equals the resonance frequency, the stiffness of the air gap balances out the acoustic mass, resulting in sound absorption peaks.

The two studies converge on the innovative use of 3D printing technology for creating perforated ACA-Meta, notably enhanced by backing them with porous materials. The first study studied the sound absorption coefficient of 3D-printed perforated ACA-Meta backed by porous materials. Adjustments in the perforation ratio of these polymer-perforated ACA-Meta demonstrate the feasibility of achieving high absorption peaks, underscoring a novel approach in acoustic absorber fabrication [90]. The second study expands this concept by examining a multilayer perforated ACA-Meta backed by porous layers and air gaps. The results show that the perforation ratio and air-gap depth significantly influence the resonant frequency for peak sound absorption, with the addition of porous material enhancing sound absorption across a wider bandwidth [95]. These diverse studies collectively highlight the significant potential of AM in creating highly efficient, custom-designed perforated ACA-Meta. This deeper understanding is crucial for developing and optimising sound-absorbing materials across a wide range of applications.

### 6.2. Studies on additive manufactured slotted ACA-Meta

This section summarises the studies on measuring the acoustic absorption of the additive manufactured slotted ACA-Meta. In exploring low-frequency acoustic absorption, the investigation focuses on a multi-cavity of coiled-up slots, as depicted in Figure 13a. At low frequencies (100–600 Hz), the acoustic metamaterial demonstrated good sound absorption; its behaviour was mostly determined by the length of the structure and the width of the slot. It is also evident that resonance frequencies and SAC are influenced by the number of slots, as seen in Figure 14a. The multi-slot absorber is a great option for regulating sound energy in relation to low frequencies because of its performance and adaptable design [96]. However, the absorption spectrum of the conventional coiled-up slot meta-structure reveals fewer peaks and narrower bandwidths, primarily because of significant sound wave reflection. The coiled-up slots increase the mismatched impedance at the interface and boost the effective parameters of the metamaterial. Hence, impedance-matching layers with variable slot distances, as shown in Figure 13b, must be introduced for effective absorption. Broadband absorption performance is demonstrated for metamaterials with impedance-matching layers when the slot distance decreases, as seen in Figure 14c. This opens up the possibility of using meta-absorber-based acoustic absorbers with suggested impedance-matching layer designs for highly effective absorption over a wide bandwidth [97].

Based on the findings by Mu et al., a resonator positioned within a hexagonal recess, as shown in Figure 13d, introduces local peaks in the frequency ranges of 1,100 Hz – 2,100 Hz and 3,600–4,600 Hz. While the peak between 3,500 and 4,500 Hz is more susceptible to resonator length fluctuations, the peak between 1,000 and 3,000 Hz is less so, as seen in Figure 14b [100]. Another study introduces a hexagonal acoustic metamaterial cell with a configurable perforating rate for numerous parallel-connection resonators. The cell has six triangular slots and six trapezium slots, as shown in Figure 13d, with perforation rates changeable by moving the sliding block along the slideway. The average sound absorption coefficients were 0.85, 0.86, and 0.88 in the target frequency ranges of 600–1200 Hz, 750–1250 Hz, and 750–1100 Hz, respectively. The adjustable sound absorption performance, excellent low-frequency noise reduction property, expandable outline construction, and effective space usage make it suitable for practical noise reduction applications [99]. Furthermore, the study presents the design of a micro-helix metamaterial, as shown in Figure 13e, that supports strong sound absorption qualities through 3D printing. Measurements from the experiment demonstrate that the sound absorption coefficient is highly dependent on the geometrical dimensions of the slot depth and the helix vestibule [101].

Fractal acoustic metamaterials may be used to build accurate tortuous path lengths and depths to control acoustic absorption across a wide frequency range. The maximum absorption values for a certain fractal order correspond to particle air velocity fields inside convoluted paths. Because fractal Hilbert geometry is self-filling, it is the best choice for producing space-

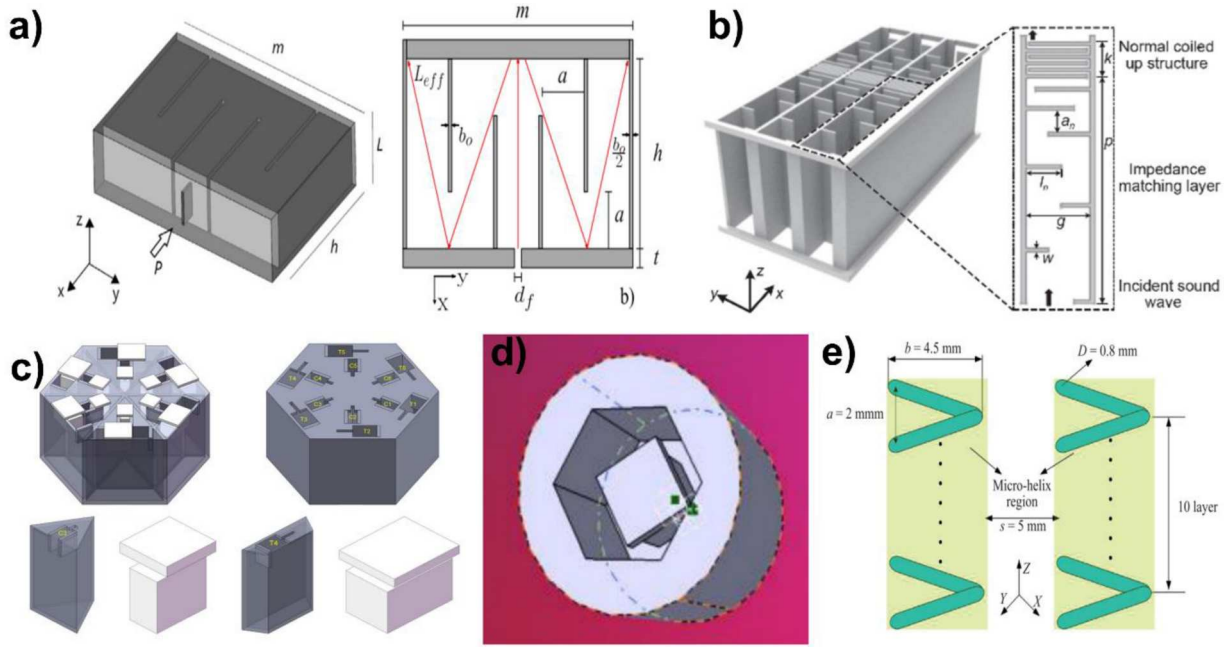



**Figure 13.** ACA-Meta with varying designs of slot structures: a,b) coil [96], Reproduced with permission from Applied Acoustics. Rights obtained from Elsevier, [98] Reproduced with permission from AIP Publishing, under the Creative Commons Attribution License (CC BY), c) triangle, trapezium [99] Reproduced with permission from Materials, under the Creative Commons Attribution License (CC BY), d) square within a hexagon [100], Reproduced with permission from ASEAN Eng. Journal, under the Creative Commons Attribution License (CC BY), and e) helical [101] Reproduced with permission from Theoretical and Applied Mechanics Letters, under the Creative Commons Attribution License (CC BY).

optimised fractal geometries. The absorption coefficient behaviour of fractal Hilbert and coiled geometry is not notably different [98].

Another study proposes a design based on individual unit cells comprising multiple resonators with a labyrinthine-coiled slot configuration for expanding the working bandwidth and downscaling the device. The proposed absorber features a planar profile, broad bandwidth, wide absorbing angle, and thin thickness [102]. Extending the exploration, a different study incorporates lossy porous media around single-resonance Helmholtz resonators (slots) necks to achieve broadband perfect acoustic absorption through damped resonances, resulting in wider absorption bandwidths [103].

### 6.3. Studies on additive manufactured cellular ACA-Meta

This section summarises the studies conducted on measuring the acoustic absorption of the additive-manufactured cellular ACA-Meta, as shown in Figure 15.

#### 6.3.1. Honeycombs

Much research has been conducted on honeycombs in terms of cellular structures. In AM, recent research has examined the interplay of sample height, density, and sample orientation in determining the acoustic absorption of manufactured honeycomb structures. This research highlights the impact of sample height and infill density, showing that the thin absorber with the back air gap absorbs sound in the mid-frequency band, similar to a Helmholtz resonator, as seen in Figure 16b. The thicker absorber may be managed by lowering the infill density to increase porosity, but this must be done carefully to avoid worsening in low to mid-frequency absorption [108]. Another research study explored the manipulation of material density, resulting in layered porous materials with varied pore diameters. By adjusting the combination of infill percentages, the sound absorption coefficient of these materials may be tuned to the appropriate frequency range. The fabricated samples have a wide frequency range of sound absorption coefficients, which is ideal for acoustic materials. By varying the amount of material filling along the wave propagation direction, the produced structures may be adjusted to the appropriate frequency range [109].

To provide useful insights into the design and optimisation of acoustic absorbers, these subsequent works collectively investigate how topology impacts acoustic absorption in additively manufactured honeycomb structures. Absorber densities, influenced by topologies like squares, rectangles, cubes, hexagons, and triangles,

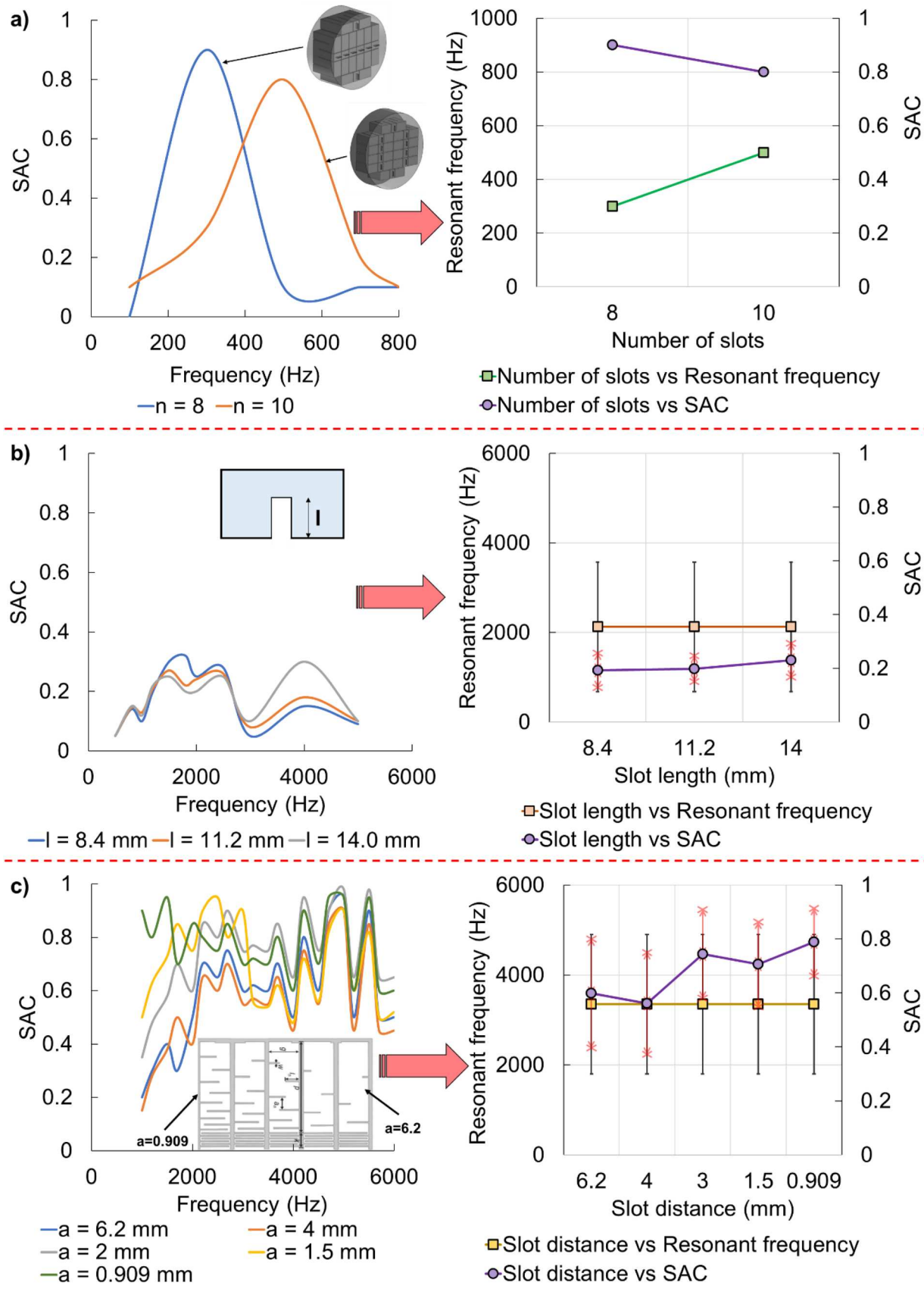



**Figure 14.** Effect of design parameters of the slotted ACA-Meta: a) number of slots [96], b) slot length [100], and c) slot distance [97] on SAC along the frequency (left) and resonant frequency, SAC (right).

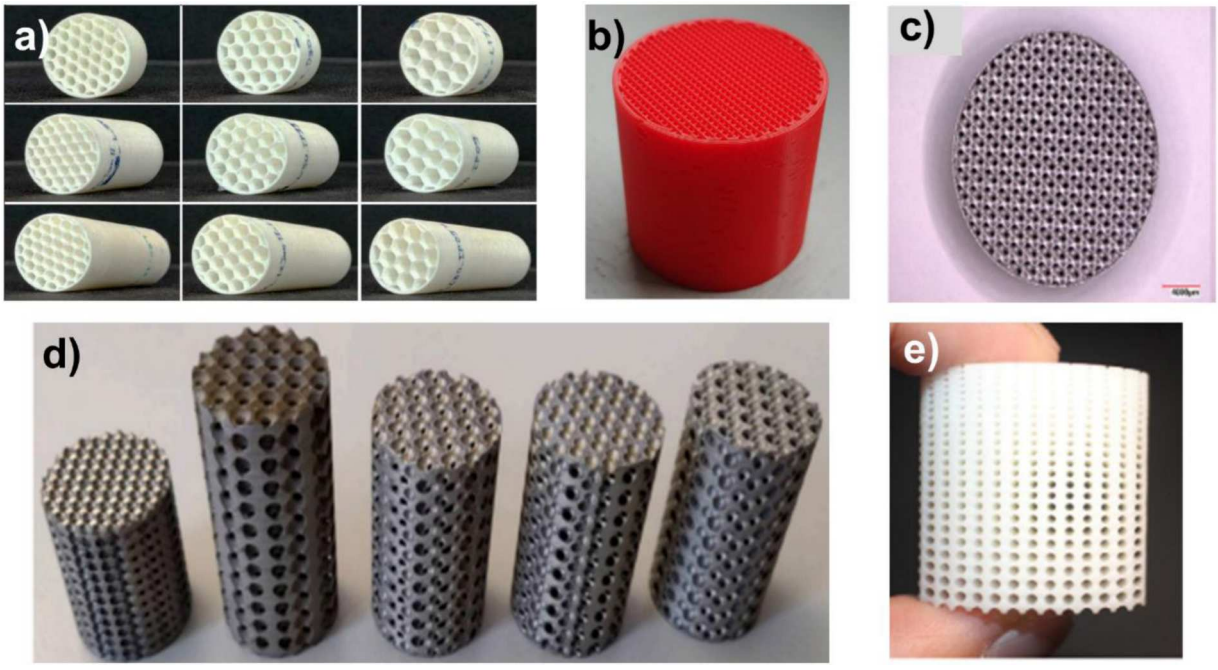



**Figure 15.** ACA-Meta with varying cellular types: a, b) honeycombs-based [104,105] Reproduced with permission from Applied Acoustics. Rights obtained from Elsevier, c) sheet/shell-based [106] Reproduced with permission from Coatings, under the Creative Commons Attribution License (CC BY), d) skeletal/ligament-based [64], Reproduced with permission from Additive Manufacturing. Rights obtained from Elsevier, e) strut-based [107] Reproduced with permission from Applied Acoustics. Rights obtained from Elsevier.

demonstrate that larger pore widths lead to reduced density, which in turn shifts absorption peaks towards the high-frequency spectrum, with further reduction moving these peaks to the low-frequency band as seen in Figure 16ai [110]. In a detailed examination of cartesian, starlit, and octagonal topologies, the starlit configuration emerges as the most effective, achieving the maximum sound absorption coefficient. This efficiency is attributed to the interplay between pore sizes and sample height, where increased height benefits low-frequency absorption, as seen in Figure 16aii, but at the cost of higher manufacturing expenses [111]. Further, the exploration of hexagonal and octagonal narrow tubes in single and multi-periodic structures reveals that by varying the cell size and topology, the frequency and amplitude of maximum absorption can be precisely controlled, as seen in Figure 16c [104]. The study presents a novel approach to understanding the complex relationship between acoustic behaviour and the micro-scale material structure, an area previously limited in comprehension. It introduces a method for calculating acoustic material properties, specifically Biot parameters, by leveraging machine learning algorithms to analyze the micro-scale shape of materials. This method involves creating 50 test samples using material extrusion with a 0.20 mm nozzle diameter, meticulously designed to replicate microscopic geometries. The data obtained from these samples is then processed through FEM and machine learning models, which have shown considerable success in predicting Biot parameters based on the micro-scale characteristics of the samples [112,113].

However, the study uncovers a significant challenge in this process: manufacturing flaws like elliptical filament sections, filament section shrinkage, and surface rugosity. These imperfections, often overlooked, have a critical impact on acoustic predictions, leading to inaccuracies. Furthermore, the study critically evaluates the discrepancies between idealised micro-lattices and their real-world counterparts, emphasising that the neglect of these micro-geometric imperfections, such as elliptical sections, shrinkage, or micro-grooves, can substantially diminish the precision of the predictive models [105].

### *6.3.2. Strut*

Two studies on strut-based structures focused on determining the effect of surface roughness on strut-based structures manufactured using various AM techniques. Both concluded that developing strut-based structures like Kelvin cells and a structure resembling interconnected cylinders using different 3D printing techniques results in imperfections on the samples owing to varying surface roughness, which has an approximate influence on the peaks of acoustic absorption [63,66]. Only one study investigated the effect of a continuously

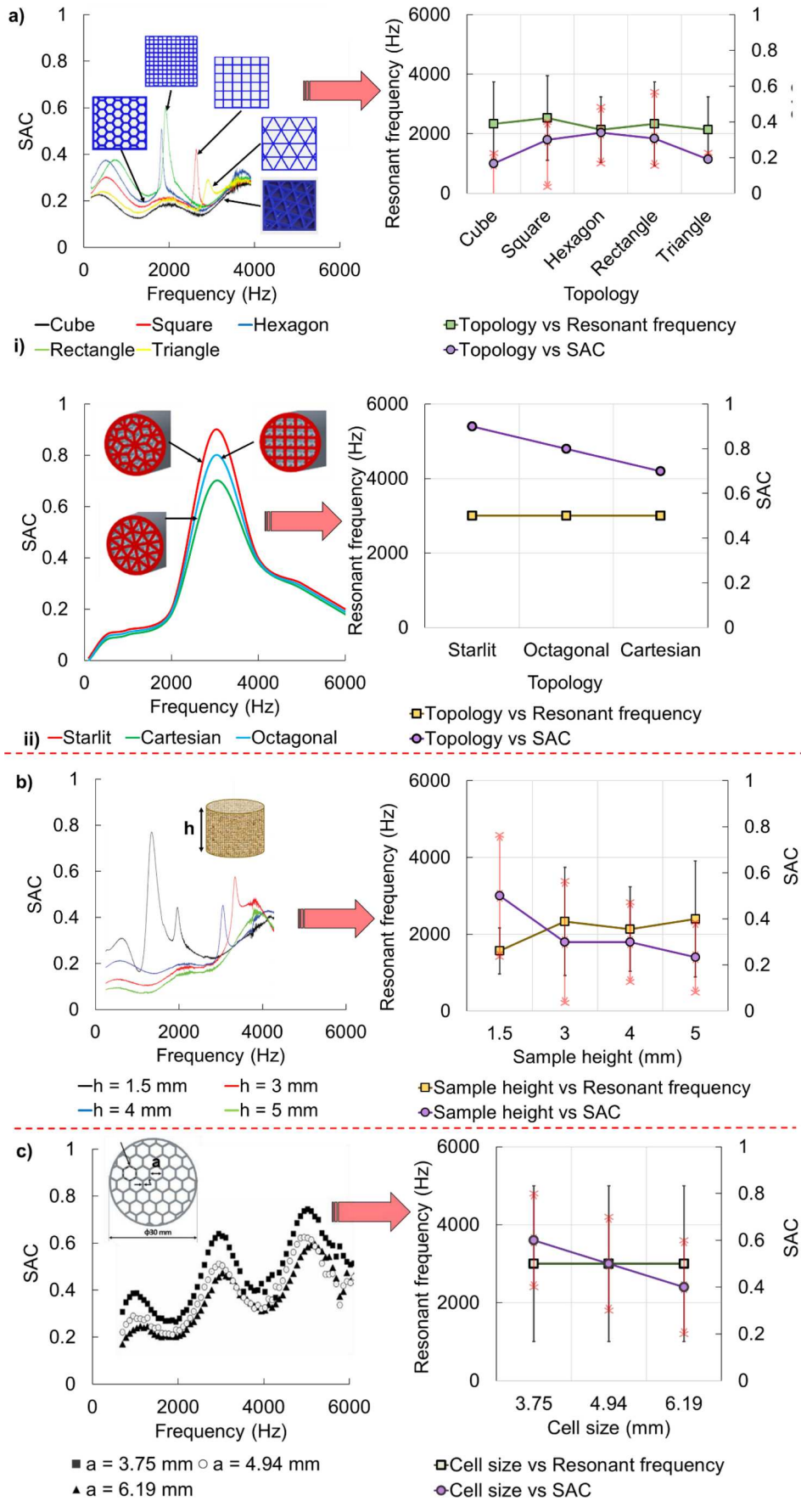



**Figure 16.** Effect of design parameters of the honeycomb cellular ACA-Meta: a) topology: i) [110], ii) [111], b) sample height [108], and cell size [104] on SAC along the frequency (left) and resonant frequency, SAC (right).

graded strut-based structure on its sound absorption ability. It finds that the structures with graded cells have better sound absorption performance because of a drop in primary reflected energy and increased absorbed energy compared to that of uniform structures [107].

#### 6.3.3. Shell/sheet

This research examines the acoustic performance of polymer hierarchical shell-based structures created with material jetting technology. Four distinct single-cell geometries were evaluated as basis cases: sphere with truncated ends, Schwartz-D, diamond, and gyroid. The Sphere sample absorbed the most sound, with a potential maximum value of around 2400 Hz. The diamond structure exhibited little porosity, allowing sound to flow through unhindered. Around 3200 Hz, the absorption coefficient climbed to 0.55 for higher frequencies. The absorption behaviour of the Schwartz-D and gyroid samples was comparable, with a maximum value of 0.56-0.61 about 1650 Hz, as seen in Figure 17ai. However, their thinness makes sound absorption in the low-frequency band insufficient for real-world applications [114]. This work evaluates the geometry-related parameters (height, unit cell size, and volume percentage) of polymeric cellular structures based on triply periodic minimum surfaces (primitive, gyroid, and diamond) for upper midrange sound absorption. The findings demonstrate that, over a broad bandwidth, the diamond surfaces had the best absorption capacities of the three varieties, as seen in Figure 17aii. A small unit cell size can provide high absorption coefficients, as seen in Figure 17bii, and sample height can change the effective frequency ranges, as seen in Figure 17cii [115]. The same trend has been seen for gyroid structure while evaluating the influence of cell size, as seen in Figure bi, and sample height, as seen in Figure 17ci, on acoustic absorption. Increasing the sample height narrows the peaks of acoustic absorption in both cases. The same study also examines the shell thickness and graded gyroid structures for sound absorption results. According to the findings, the resonance frequency of the gyroid is not affected by the TPMS surface thickness. However, its sound absorption coefficient increases with shell thickness, as seen in Figure 17d, and graded porosity increases the sound absorption capacity of TPMS structures substantially. The sound wave enters the porous cavity from a tiny porous structure, and its element size rises from 2 mm to 4 mm, which is the process of sound absorption in graded TPMS. When the sound wave hits the barrier and returns, it scatters several times, resulting in a low-frequency resonance. Surface porosity at the contact can be reduced by increasing internal porosity and improving sound absorption [106].

#### 6.3.4. Skeletal/ligament

This study focuses on the applicability of AM technologies like FDM, SLA, SLS, SLM, and DLP for reproducing skeletal-based sound absorption samples. Independent manufacturing and acoustic measurements were performed by various laboratories using the same geometry codes, representing accepted periodic cellular patterns. The findings obtained from measurements done on samples of the same design are fairly similar; however, some variations are attributable to manufacturing process-induced shape and surface flaws or microporosity. Overall manufacturing quality linked with the AM technique and the material used for production, tiny flaws arising during the production process, and post-production treatment are three explanations for variations between measurement findings [64]. A similar kind of study was conducted by comparing AM techniques like SLA and CJP, providing a similar observation type [65]. Only in the case of skeletal structures have non-periodic types been explored. Spinodoid structures, or spinodoid metamaterials, are non-periodic cellular structures that replicate spinodal topologies found during diffusion-driven phase separation processes. These structures, which may be made using additive methods, provide an appealing approach to constructing multifunctional structures. The normal incidence sound absorption behaviour of four various spinodoid topologies is investigated in this paper: isotropic, cubic, columnar, and lamellar. According to the findings, the open pore network and acoustical features of spinodoid structures are predominantly determined by their relative density and wavenumber. Lamellar spinodoids have distinct low-frequency sound absorption behaviour, but isotropic, cubic, and columnar spinodoids exhibit comparable behaviour [41].

### 6.4. Studies on additive manufactured hybrid ACA-Meta

This section summarises the studies conducted on measuring the acoustic absorption of the additive-manufactured hybrid ACA-Meta. Recent advancements in acoustic metamaterials have seen a remarkable fusion of diverse structures which are designated as hybrid in this review, such as Hilbert fractals, coiled designs, perforated panels, and lattice configurations, each contributing uniquely to the enhancement of sound absorption, particularly in low-frequency domains. Metamaterials that merge Hilbert fractals with coiled structures utilise multi-order resonance modes to attenuate low-

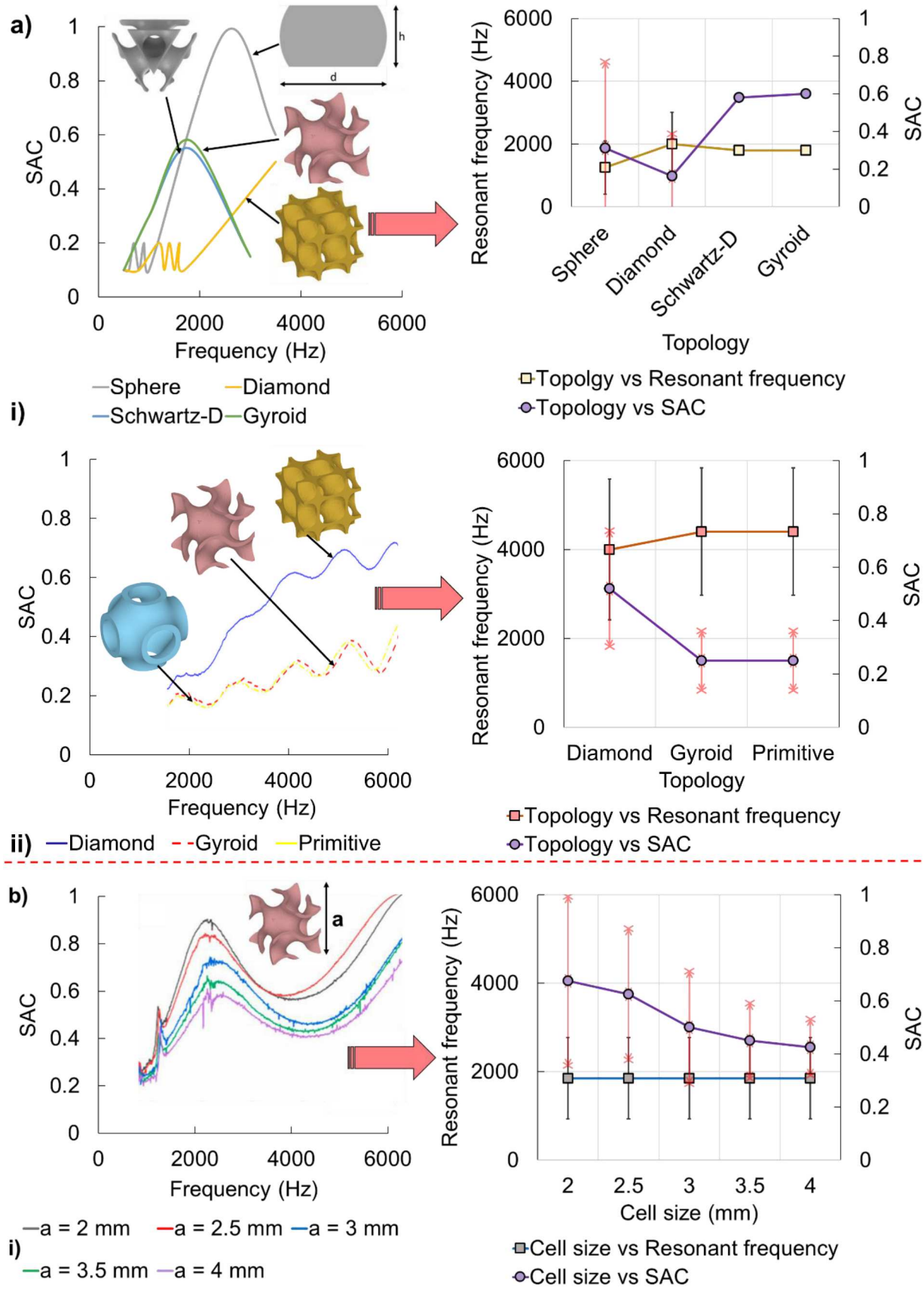



**Figure 17.** Effect of design parameters of the sheet/shell cellular ACA-Meta: a) topology: i) h = 20 mm [114], ii) h = 11.2 mm, b) cell size: i) gyroid [106], ii) diamond [115], sample height: gyroid [106], ii) diamond [115], and shell thickness [106] on SAC along the frequency (left) and resonant frequency, SAC (right).

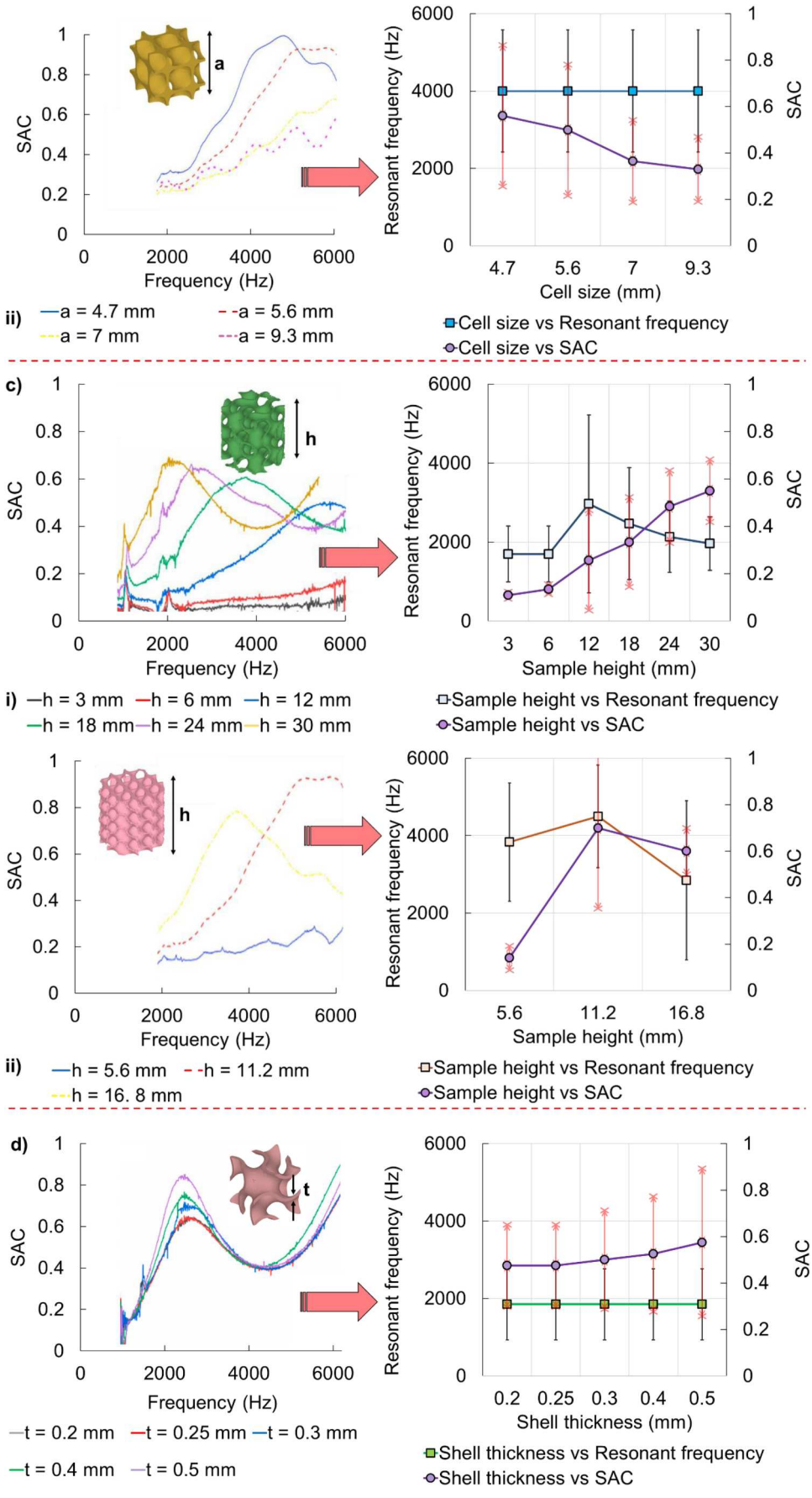



**Figure 17.** *Continued*

frequency noise [106] effectively. Innovations extend to hollow truss-plate hybrids with micro-perforations, as seen in Figure 18c, leading to the development of micro-lattice metamaterials characterised by their quasi-perfect sound absorption and broadband capabilities [116]. The integration of perforated struts in these metamaterials not only boosts sound absorption but also enhances mechanical efficiency, with 3D printing techniques enabling the creation of structures adept at broad low-frequency absorption [117]. Further, sandwich constructions with micro-perforated cores [118] and acoustic absorption metamaterials with adjustable coiled slots [119] showcase the versatility and effectiveness of these designs. The coiled Helmholtz slotted structures [120] and labyrinthine metamaterials with internally folded slots [121] stand out for their unique wave manipulation and reflection approaches at band gap frequencies. The field also sees novel designs like perforated honeycomb-corrugation hybrids [27], segmented membrane sound absorbers [122], spirally slotted-perforated structures [80], and meta-silencers as seen in Figure 18d, with customisable intrinsic loss and mode density [123], each opening new avenues in acoustic material science. Hierarchical metamaterials, as seen in Figure 18a [124], integrated passive devices for aerospace applications [125], resonator-based periodic structures [126], and perforated honeycombs with varied infill densities [127] further illustrate the breadth of research in this domain. These developments, including series-parallel organisation structures as seen in Figure 18b [128] and honeycomb hexagonal Helmholtz resonator slots [129], span a wide frequency range for effective noise attenuation and show potential for diverse industrial applications. Innovations like asymmetric absorption cells [130], perforated with zig-zag slots [131], and studies on manufacturing solutions, such as extrusion printing with freeze casting for honeycombs with perforated plates, as seen in Figure 18e [132], underscore the continual evolution and potential of acoustic metamaterials in addressing complex noise control challenges and offering novel solutions in acoustic engineering.

## 7. Summary and perspective

This section summarises the investigations on acoustic metamaterials based on fabrication, materials, and applications, as shown in Figure 19. Table 2 summarises all the data, considering the above-mentioned factors. Refer to Figure 10 for the listed parameters.

### 7.1. Design perspective

This section highlights the designs explored and presents potential concepts for future developments. Studies on additively manufactured perforated ACA-Meta have demonstrated that adjusting the perforation ratio alters the resonance frequency of the peak sound absorption coefficient, with higher ratios increasing the frequency. The influence of perforation volume, including thickness, diameter, and material density, significantly impacts sound absorption, resonant frequencies, and bandwidth. For instance, 3D printing infill density affects the mass and resonance of 3D-printed specimens, where denser materials maintain higher SAC without changing the frequency spectrum. Furthermore, geometric aspects such as porosity, hole diameter, sample height, and perforation angles play vital roles, with height notably affecting acoustic behaviour. Slotted ACA-Meta improves low-frequency acoustic absorption. Novel configurations, such as hexagonal cells with parallel-connected resonators, micro-helix designs, and fractal-inspired geometries, demonstrate tunable absorption properties across wide frequency ranges. Regarding cellular ACA-Meta, honeycomb structures and strut-based configurations underscore the impact of topology, density, height, and manufacturing imperfections on acoustic performance. Shell-based structures like Schwartz-D, gyroid, and sphere exhibit unique frequency-dependent absorption behaviours. Figure 19c clearly demonstrates that the ACA-metamaterials exhibit a satisfactory SAC across different frequency ranges. For instance, the perforated ACA-Meta is suitable for mid-frequency applications, the slotted version performs well in low to mid-frequency ranges, and the cellular ACA-Meta can be applied across the entire spectrum when carefully optimised. In addition to the configurations detailed in this study, other configurations inspired by nature, biology, and mechanisms have also been explored to an extent. There is significant potential to advance the design of periodic cellular ACA-Meta by grading parameters such as density, cell size, and topology. Additionally, further research is required to thoroughly understand the acoustic properties of non-periodic cellular ACA-Meta. Moreover, exploring hybrid ACA-Meta offers numerous possible combinations, presenting an exciting opportunity for future investigations.

### 7.2. Additive manufacturing perspective

This section highlights the materials and AM methods explored and presents potential advancements for future developments. By selecting the proper material and AM method, it is evident that these metamaterials possess exceptional mechanical strength, thermal stability, environmental resistance, and other critical properties (see e.g. [133,134]) required for acoustic

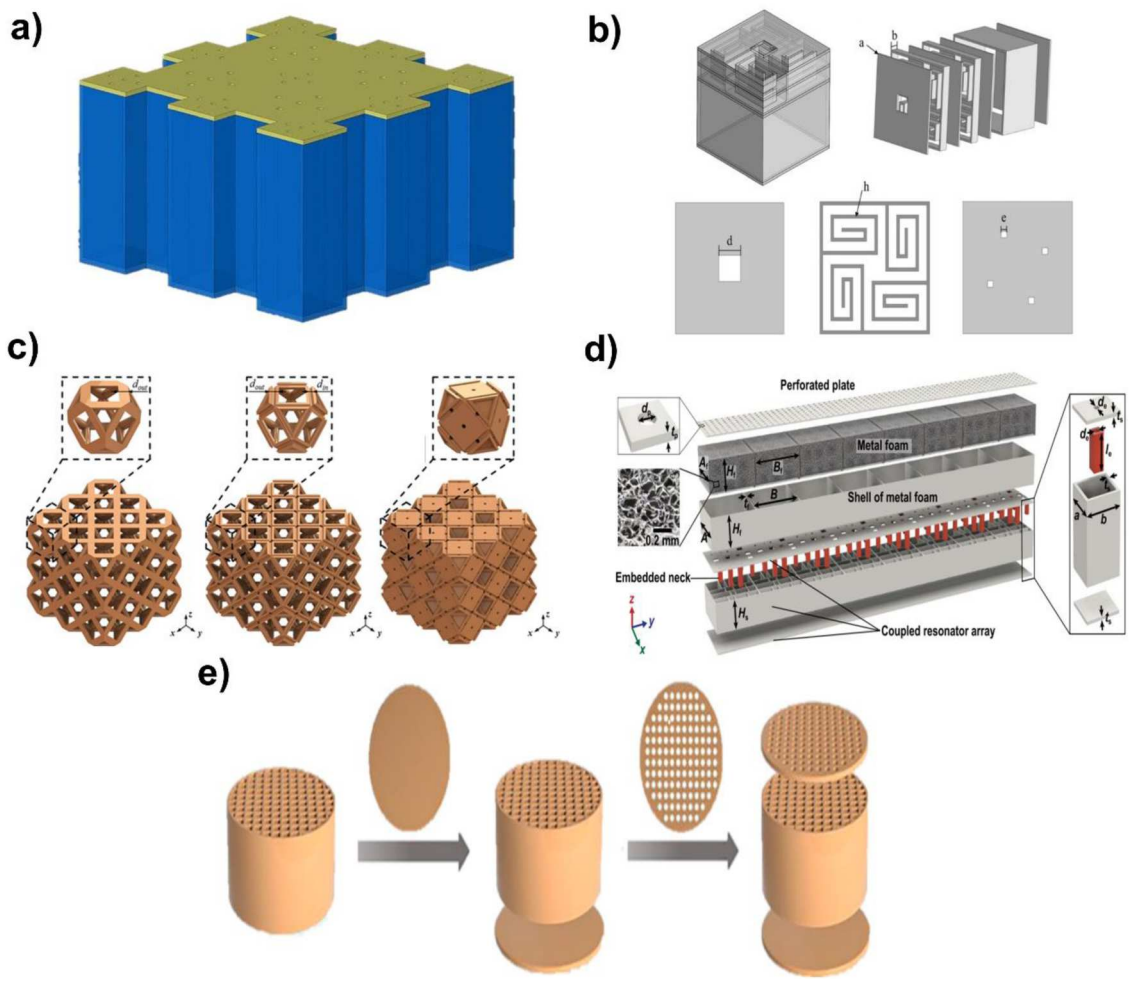



**Figure 18.** ACA-Meta with various hybrid combinations: a) hierarchically perforated-slotted [124] Reproduced with permission from Materials & Design, under the Creative Commons Attribution License (CC BY), b) hierarchically slotted-perforated [128] Reproduced with permission from Journal of Low Frequency Noise, Vibration and Active Control, under the Creative Commons Attribution License (CC BY), c) cellular (strut)-perforated [116] Reproduced with permission from International Journal of Mechanical Sciences. Rights obtained from Elsevier, d) perforated-cellular (foam)-slotted [123] Reproduced with permission from IOP Publishing Ltd, under the Creative Commons Attribution License (CC BY), and e) cellular (honeycombs)-perforated [132] Reproduced with permission from Additive Manufacturing. Rights obtained from Elsevier.

applications. Polymer-based materials have been the main emphasis from a materials perspective because of their cost-effectiveness and ubiquitous availability. Photopolymer resins have also been utilised; however, they are more expensive than polymer filaments. Relatively limited research has been done on materials based on metals and other materials like aerogels and powders like calcium sulphate. From the methodology perspective, most research on this subject has focused on extrusion technologies, owing to their economical nature. Furthermore, few studies have investigated vat photopolymerization, drawn by its potential ability to offer precise surface finishes while maintaining tolerances below acceptable limits. It is also worth mentioning that imperfections in other processes, such as material jetting, binder jetting, and powder bed fusion, can also contribute to acoustic absorption. While only a few studies have looked into this issue, further studies are needed to evaluate their effectiveness fully. Remarkably, no investigations have been published on fabricating acoustic metamaterials employing sheet lamination and direct energy deposition techniques. The dearth of study in these areas is most likely due to the challenges these technologies encounter when creating complicated structures, a fundamental prerequisite for developing acoustic metamaterials. To overcome these problems, creative solutions and concerted research efforts in sheet lamination and direct energy deposition will be required. Additionally, one of the limitations of AM is the challenge of scaling up to

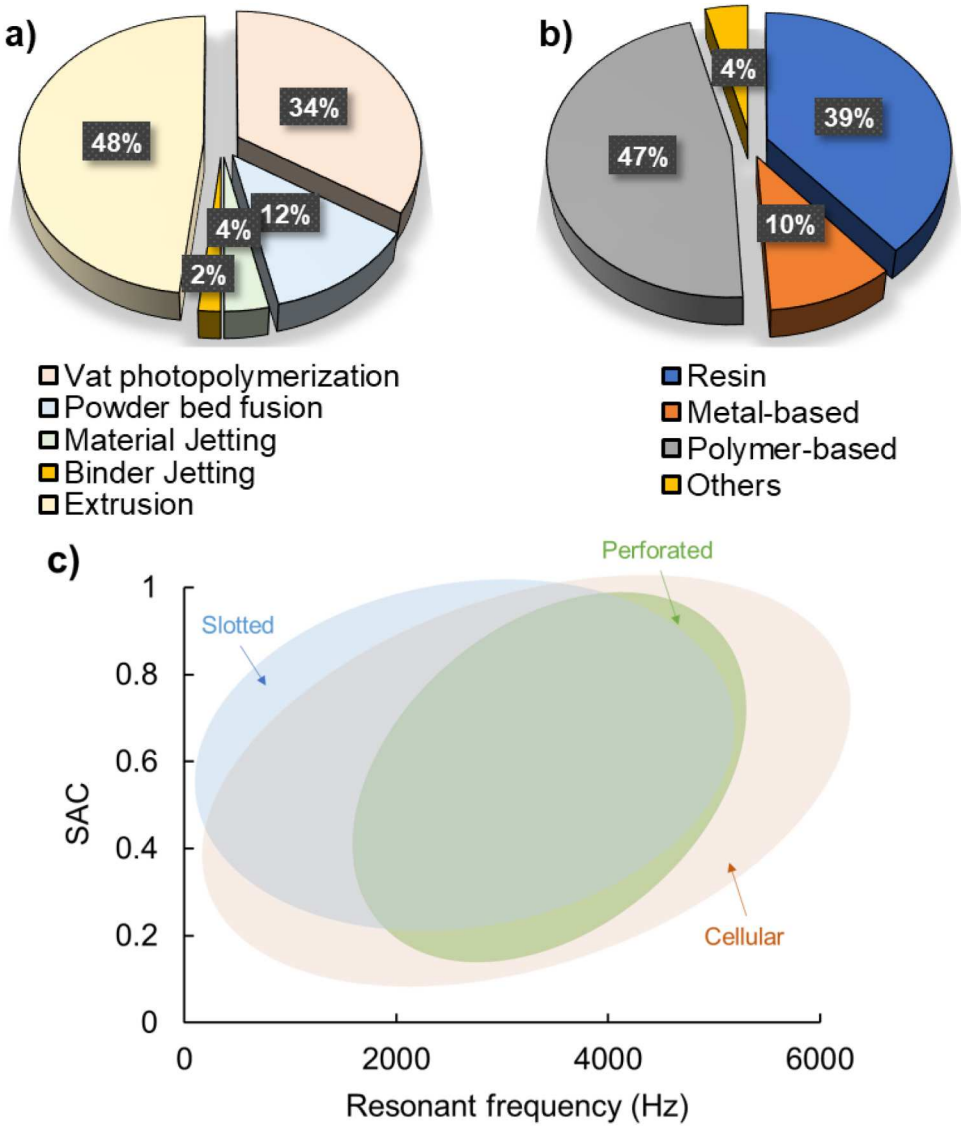



**Figure 19.** Summary of ACA-Meta on a) fabrication techniques, b) materials, and c) application based on frequency ranges. Data taken from [41,63–66,69,86–115].

produce large acoustic panels. Large-scale printing frequently faces problems such as maintaining material consistency, ensuring dimensional accuracy, and extended production times, all of which impede the broader adoption of these technologies for manufacturing full-sized acoustic solutions. Furthermore, the high installation costs of 3D printers and the limited range of materials available for printing add to the difficulties in large-scale fabrication [135].

In the near future, advances in AM are expected to revolutionise the production of acoustic metamaterials through several ground-breaking developments. One promising direction is the development of micro-architected metamaterials using micro/nano-3D printing technologies [136]. This approach enables the precise fabrication of highly intricate designs, allowing much smaller geometric features and ultralightweight ACA-Meta. However, the main issue with micro/nano-3D printing techniques is scalability, where there is a large need for increasing printing speeds. One promising approach addressing speed is the recently developed volumetric AM technique [137]. Furthermore, AM techniques such as multi-laser beam systems in laser powder-bed fusion, combined with the capability to produce exceptionally large build volumes spanning several metres, promise to transform the industrial fabrication of acoustic metamaterials [138]. Additionally, advances in 4D printing bring a new dimension to this field by incorporating smart materials and structures. The ability to print with shape-memory alloys or polymers introduces dynamic functionality, making it possible to develop adaptable and reconfigurable acoustic

**Table 2.** Summary data of the additive manufactured ACA-Meta [41,63–66,69,86–132].

| S.No | Type of ACA-Meta | Methods | Techniques | Materials | Frequency range investigated | Parameters / combinations investigated* |
|---|---|---|---|---|---|---|
| 1. | Perforated | VPP | SLA | VisiJet-SL | 500 Hz to 5500 Hz | b, D′ |
| 2. | Perforated | VPP | SLA | VisiJet-SL | 500 Hz to 6000 Hz | θ, D′ |
| 3. | Perforated | VPP | SLA | VisiJet-SL | 400 Hz to 6400 Hz | D′ |
| 4. | Perforated | VPP | SLA | Resin | 100 Hz to 500 Hz | Panel arrangement |
| 5. | Perforated | VPP | SLA | Resin | 250 Hz to 2250 Hz | D′ |
| 6. | Perforated | PBF | DMLS | Titanium | 400 Hz to 1600 Hz | Perforation geometries |
| 7. | Perforated | PBF | SLS | Polyamide | 200 Hz to 1600 Hz | Panel arrangement |
| 8. | Perforated | MJ | MJP | VisiJet | 0 Hz to 6000 Hz | Panel porosity, h, d |
| 9. | Perforated | Extrusion | FDM | PLA-wood | 500 Hz to 4500 Hz | Perforation volume |
| 10. | Perforated | Extrusion | FDM | PLA-wood | 500 Hz to 4500 Hz | Panel density |
| 11. | Perforated | Extrusion | FDM | PLA | 200 Hz to 2000Hz | Perforation cross-sections |
| 12. | Slotted | Extrusion | FDM | ABS | 100 Hz to 800 Hz | l, e |
| 13. | Slotted | Extrusion | FDM | PLA | 1000 Hz to 6000 Hz | c |
| 14. | Slotted | Extrusion | FDM | PLA | 500 Hz to 5000 Hz | l |
| 15. | Slotted | Extrusion | FDM | - | 200 Hz to 1600 Hz | Slot shape |
| 16. | Slotted | Extrusion | FDM | PLA | 200 Hz to 1600 Hz | Slot geometries |
| 17. | Slotted | Extrusion | FDM | PLA | 500 Hz to 2500 Hz | Slot shape |
| 18. | Slotted | Extrusion | FDM | - | 1300 Hz to 2100 Hz | Slot geometries |
| 19. | Cellular | VPP | SLA | Resin | 2000Hz to 6000 Hz | a, cell topology, h |
| 20. | Cellular | VPP | DLP | Resin | 500 Hz to 3500 Hz | Surface roughness |
| 21. | Cellular | VPP | SLA | Resin | 0 Hz to 6000 Hz | Surface roughness |
| 22. | Cellular | VPP | DLP | Resin | 0 Hz to 6000 Hz | Surface roughness |
| 23. | Cellular | VPP | - | Resin | 0 Hz to 6000 Hz | Surface roughness |
| 24. | Cellular | PBF | SLM | AlSi7Mg | 0 Hz to 6000 Hz | a, cell grading, h, t |
| 25. | Cellular | PBF | SLM | Cobalt chrome | 500 Hz to 3500 Hz | Surface roughness |
| 26. | Cellular | PBF | SLM | Al, CoCr, SS | 0 Hz to 6000 Hz | Surface roughness |
| 27. | Cellular | PBF | SLM | Metal powder | 500 Hz to 3500 Hz | Surface roughness |
| 28. | Cellular | PBF | SLS | Polyamide | 0 Hz to 6000 Hz | Surface roughness |
| 29. | Cellular | MJ | MJP | Resin | 100 Hz to 3000 Hz | Cell topology |
| 30. | Cellular | BJ | CJP | Calcium sulfate hemihydrate | 0 Hz to 6000 Hz | Surface roughness |
| 31. | Cellular | Extrusion | FDM | PLA | 0 Hz to 6000 Hz | k, a |
| 32. | Cellular | Extrusion | FDM | PLA-wood | 500 Hz to 4500 Hz | h, sample density |
| 33. | Cellular | Extrusion | FDM | PLA-wood | 500 Hz to 4500 Hz | Cell topology |
| 34. | Cellular | Extrusion | FDM | PLA | 0 Hz to 6000 Hz | Surface roughness |
| 35. | Cellular | Extrusion | FDM | ABS | 0 Hz to 6000 Hz | a, cell topology |
| 36. | Cellular | Extrusion | FDM | ABS | 0 Hz to 6000 Hz | Sample density |
| 37. | Cellular | Extrusion | FDM | PLA | 500 Hz to 3500 Hz | Surface roughness |
| 38. | Cellular | Extrusion | FDM | ABS | 0 Hz to 6000 Hz | Cell topology |
| 39. | Cellular | Extrusion | FDM | PLA, ABS | 0 Hz to 6000 Hz | Surface roughness |
| 40. | Cellular | Extrusion | FDM | PLA | 0 Hz to 6000 Hz | Cell topology |
| 41. | Hybrid | VPP | SLA | Resin | 30 Hz to 1600 Hz | Perforated-slotted |
| 42. | Hybrid | VPP | SLA | Resin | 500 Hz to 3200 Hz | Perforated-slotted-cellular (foam) |
| 43. | Hybrid | VPP | SLA | Resin | 200 Hz to 1600 Hz | Cellular (honeycombs)-perforated |
| 44. | Hybrid | VPP | DLP | Resin | 200 Hz to 2000Hz | Perforated-slotted |
| 45. | Hybrid | VPP | - | Resin | 0 Hz to 6000 Hz | Cellular (plate)-perforated |
| 46. | Hybrid | VPP | - | Resin | 800 Hz to 1800Hz | Perforated-slotted |
| 47. | Hybrid | VPP | DLP | Resin | 200 Hz to 1000 Hz | Cellular (strut)-slotted-perforated |
| 48. | Hybrid | VPP | DLP | Resin | 1000 Hz to 6000 Hz | Cellular (strut)-perforated |
| 49. | Hybrid | VPP | DLP | Resin | 0 Hz to 6000 Hz | Perforated-slotted |
| 50. | Hybrid | PBF | SLS | Polyamide | 100 Hz to 3000 Hz | Spirally slotted-perforated |
| 51. | Hybrid | Extrusion | FDM | PLA | 30 Hz to 1600 Hz | Perforated-slotted |
| 52. | Hybrid | Extrusion | FDM | PLA | 300 Hz to 800 Hz | Perforated-slotted |
| 53. | Hybrid | Extrusion | FDM | - | 500 Hz to 3500 Hz | Perforated-slotted |
| 54. | Hybrid | Extrusion | FDM | ABS | 0 Hz to 1500 Hz | Perforated-slotted |
| 55. | Hybrid | Extrusion | FDM | PLA | 500 Hz to 3000 Hz | Cellular (honeycombs)-perforated |
| 56. | Hybrid | Extrusion | FDM | ABS | 0 Hz to 6000 Hz | Perforated-slotted |
| 57. | Hybrid | Extrusion | DIW | Aerogel | 0 Hz to 6000 Hz | Perforated-slotted |

*sample height (h), perforation diameter (d), distance between the perforations (b), perforation angle (θ) separated by an air gap (D′) to the rigid wall, slot length (l), slot thickness (e), slot spacing (c), cell size (a), cell spacing (k), shell thickness (t).

systems that respond to environmental stimuli [139]. Moreover, hybrid 3D printing is set to further enhance the field by leveraging metal laser powder-bed fusion to enable multi-material printing with functional grading [140]. These breakthroughs enable the seamless integration of materials with specific qualities inside a single structure, optimising their performance for complex and diverse acoustic applications. These breakthroughs in AM together outline a disruptive path, opening up new possibilities for constructing acoustic metamaterials with unmatched accuracy, scalability, and multifunctionality while meeting the rising need

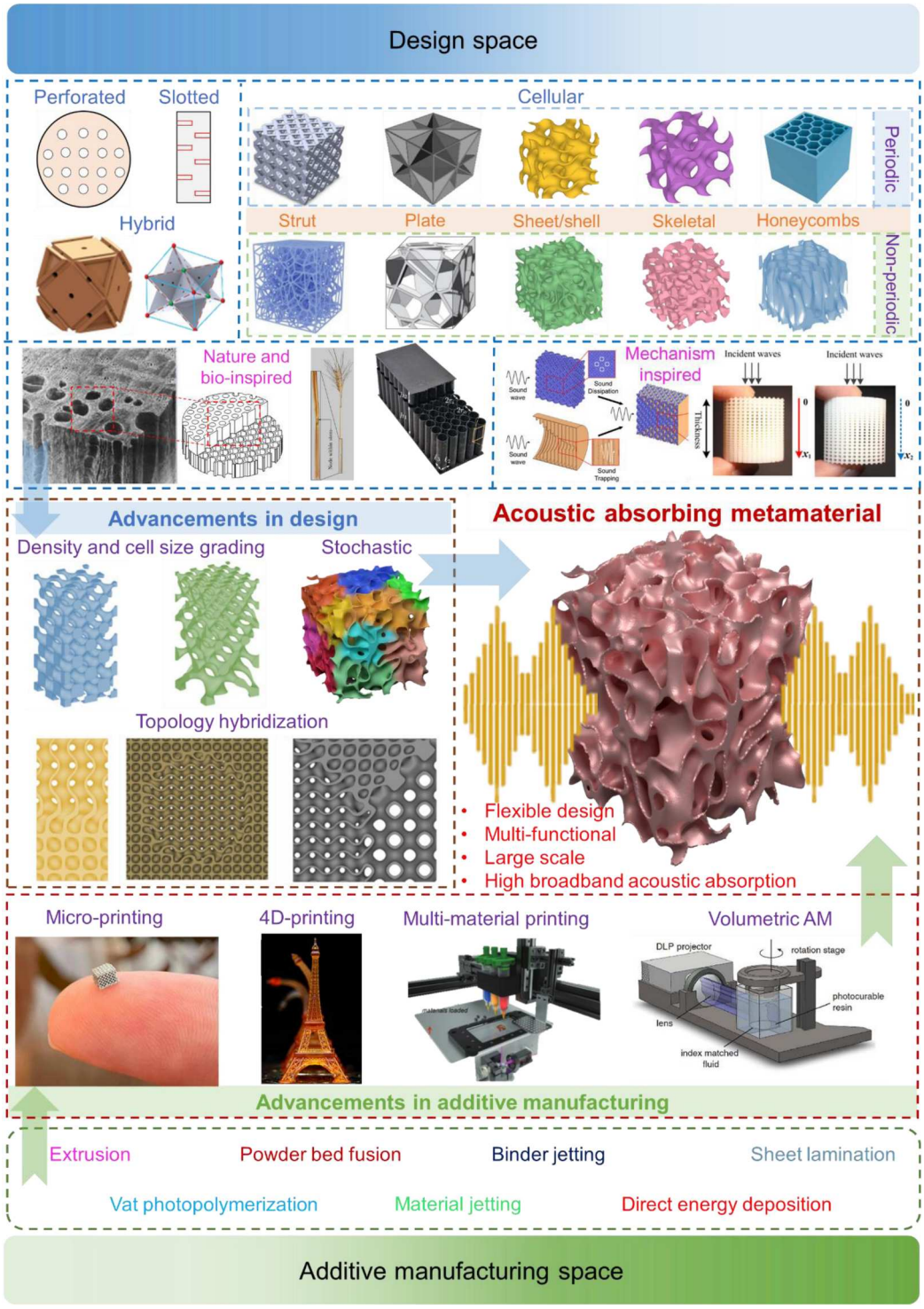



**Figure 20.** Highly efficient acoustic absorbing metamaterial with the possibility of exploring design (perforated, slotted, hybrid [116] Reproduced with permission from International Journal of Mechanical Sciences. Rights obtained from Elsevier, [118] Reproduced with permission from Materials and Design, under the Creative Commons Attribution License (CC BY), nature-, bio – [141] Reproduced with permission from Materials, under the Creative Commons Attribution License (CC BY), [142] Reproduced with permission from Scientific Reports, under the Creative Commons Attribution License (CC BY), and mechanism-inspired (sonic black hole) [143] Reproduced with permission from Composite Structures. Rights obtained from Elsevier, (impedance matching) [107] Reproduced with permission from Applied Acoustics. Rights obtained from Elsevier), advancements in design (density grading, cell size grading, stochastic, and topology hybridisation), manufacturing techniques, and advancements in manufacturing techniques such as micro-printing [144] Reproduced with permission from Materials Letters. Rights obtained from Elsevier, 4D-printing [145] Reproduced with permission from Scientific Reports, under the Creative Commons Attribution License (CC BY), multi-material printing [146] Reproduced with permission from Nano. Micro Small, under the Creative Commons Attribution License (CC BY), and volumetric AM [137] Reproduced with permission from Progress in Polymer Science. Rights obtained from Elsevier.

for improved acoustic solutions in various sectors. Figure 20 summarises the design and additive manufacturing space that has been discussed under Section 7.1 and Section 7.2.

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## Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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